2D Vectors

Questions 1–44

Question 1. It is given that A and B have position vectors $\begin{pmatrix}2\\6\end{pmatrix}$ and $\begin{pmatrix}-3\\8\end{pmatrix}$ respectively.

a) Find $\overrightarrow{AB}$.
b) Find $|\overrightarrow{AB}|$.
c) Find a unit vector in the direction of $\overrightarrow{AB}$.

Question 2. It is given that P and Q have position vectors $4\mathbf{i}-\mathbf{j}$ and $7\mathbf{i}+8\mathbf{j}$ respectively. Find a vector of length $6\sqrt{10}$ in the direction of $\overrightarrow{PQ}$.

Question 3. It is given that S and T have position vectors $\begin{pmatrix}\sqrt{150}\\ \sqrt{162}\end{pmatrix}$ and $\begin{pmatrix}\sqrt{96}\\ \sqrt{98}\end{pmatrix}$ respectively. Find a vector of length $\sqrt{126}$ in the direction of $\overrightarrow{ST}$.

Question 4. It is given that A and B have position vectors $(k-6)\mathbf{i}+(2k+4)\mathbf{j}$ and $(2k-3)\mathbf{i}+(3k+1)\mathbf{j}$ respectively, where k is a constant.

a) The vector $\mathbf{p}$ is of length $\sqrt{4k^2+36}$ in the direction of $\overrightarrow{AB}$. Find the vector $\mathbf{p}$.

b) Given that k is a positive integer, find the shortest possible vector $\mathbf{p}$.

Question 5. Find the scalar product of

a) $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

b) $\mathbf{a} = (4\mathbf{i} + \mathbf{j})$ and $\mathbf{b} = (-2\mathbf{i} + 5\mathbf{j})$

c) $\mathbf{a} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ -6 \end{pmatrix}$

d) $\mathbf{a} = 9\mathbf{j}$ and $\mathbf{b} = 6\mathbf{i} - \mathbf{j}$

Question 6. Find the value of $(3p\mathbf{i} - 2\mathbf{j}) \cdot (-5\mathbf{i} - 4p\mathbf{j})$ in terms of p.

Question 7. Given that q is nonzero, find the value of $\left(8q^3\mathbf{i} + \frac{5}{q}\mathbf{j}\right) \cdot \left(-\frac{3}{q^2}\mathbf{i} + 9q^2\mathbf{j}\right)$ in terms of q.

Question 8. Use the scalar product to find the angle between the following pairs of vectors. Give your answer in degrees to 1 decimal place.

a) $\mathbf{a} = \begin{pmatrix} 1 \\ 9 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

b) $\mathbf{a} = -2\mathbf{i} + 10\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} + 9\mathbf{j}$

c) $\mathbf{a} = \begin{pmatrix} -2 \\ -11 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 10 \\ 3 \end{pmatrix}$

d) $\mathbf{a} = -8\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = -10\mathbf{i} + 2\mathbf{j}$

Question 9. It is given that $A(2,4)$, $B(-2,1)$ and $C(-1,5)$. Find the angle ABC to 1 decimal place.

Question 10. Given that the vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, state the value of $\mathbf{a} \cdot \mathbf{b}$.

Question 11. In each case, determine if the following pairs of vectors are perpendicular.

a) $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$

b) $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -2\mathbf{i} + 5\mathbf{j}$

c) $\mathbf{a} = \begin{pmatrix} -8 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$

d) $\mathbf{a} = \sqrt{8}\mathbf{i} - \sqrt{32}\mathbf{j}$ and $\mathbf{b} = \sqrt{72}\mathbf{i} + \sqrt{18}\mathbf{j}$

Question 12. Determine the value(s) of p that would make the following pairs of vectors perpendicular:

a) $\mathbf{a} = \begin{pmatrix} 2p \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$

b) $\mathbf{a} = 6\mathbf{i} - 4p\mathbf{j}$ and $\mathbf{b} = 8p\mathbf{i} + \frac{1}{2}\mathbf{j}$

c) $\mathbf{a} = \begin{pmatrix} 4+p \\ 3p \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2p \\ 1-p \end{pmatrix}$

d) $\mathbf{a} = (2p-1)\mathbf{i} + (3p-1)\mathbf{j}$ and $\mathbf{b} = (p-2)\mathbf{i} + (2p+3)\mathbf{j}$

Question 13. Given that k is a constant, show that the vectors $\mathbf{a} = \begin{pmatrix} 3k+2 \\ 2k+2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6k-6 \\ 9k+6 \end{pmatrix}$ are perpendicular for all values of k.

Question 14. Given that k is a constant, show that the vectors $\mathbf{a} = \begin{pmatrix} k+1 \\ 6k \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2-k \\ 4k+1 \end{pmatrix}$ can never be perpendicular.

Question 15. Find the vector equation of the line that:

a) passes through $(3, 1)$ and is parallel to $2\mathbf{i} + \mathbf{j}$

b) passes through $(-2, -5)$ and is parallel to $-\mathbf{i} + 9\mathbf{j}$

c) passes through $(0, -8)$ and is parallel to $\mathbf{j}$

d) passes through (a, b) and is parallel to $p\mathbf{i} + q\mathbf{j}$, where a, b, p and q are constants

Question 16. Find a vector equation of the line that:

a) passes through $(1, 4)$ and $(2, 9)$

b) passes through $(3, 5)$ and $(5, 15)$

c) passes through $(-2, 1)$ and $(0, -5)$

d) passes through $(-4, 3)$ and $(4, -13)$

e) passes through $(2, 9)$ and $(-3, -20)$

f) passes through $(-4, -1)$ and $(-10, -7)$

g) passes through $(k, k + 1)$ and $(k - 2, 2k)$, where k is a constant

Question 17. Determine whether the point $(3, 2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

Question 18. Determine whether the point $(3, -2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} -4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Question 19. Determine whether the point $(-15, 15)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

Question 20. It is given that the point $(4, -2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} -3 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} k \\ -2 \end{pmatrix}$, where k is a constant. Find the value of k.

Question 21. The point $(-3, -8)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ k \end{pmatrix}$, where k is a constant. Find the value of k.

Question 22. Determine whether the following pairs of lines are coincident, parallel or if they intersect. If they intersect, find the coordinates of the point of intersection:

a) $l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 2 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -5 \\ 21 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 5 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 9 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 25 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -18 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -6 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 6 \end{pmatrix}$

Question 23. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} -12 \\ 3 \end{pmatrix}$, and the line l_2 is parallel to the vector $5\mathbf{i} + 20\mathbf{j}$. Show that l_1 and l_2 are perpendicular.

Question 24. Determine whether the following pairs of lines are perpendicular or not:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 3 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -7 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 7 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 9 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -9 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 9 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -11 \end{pmatrix} + \lambda \begin{pmatrix} 11 \\ 3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 11 \end{pmatrix}$

Question 25. It is given that the following lines are perpendicular:

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ -100 \end{pmatrix} + \lambda \begin{pmatrix} k \\ -3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 557 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 9 \end{pmatrix},$$

where k is a constant.

a) Find the value of k.

b) Find the position vector of the point of intersection of the two lines.

Question 26. It is given that the following lines are perpendicular:

$$l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ k \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -2 \\ 14 \end{pmatrix} + \mu \begin{pmatrix} 20 \\ 20 \end{pmatrix},$$

where k is a constant.

a) Find the value of k.

b) l_1 and l_2 intersect at the point P. Find the coordinates of the point P.

c) Find a unit vector in the direction of the position vector of P.

Question 27. It is given that the following lines intersect and are not coincident:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 15 \\ k \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 2 \end{pmatrix},$$

where k is a constant. Find the possible values of k.

Question 28. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ 420 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -10 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 120 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ -3 \end{pmatrix}.$$

a) Find the coordinates of the point of intersection of the two lines.
b) It is given that the line l_3 passes through the point $(90, 100)$ and the intersection of l_1 and l_2. Find the equation of the line l_3.

Question 29. Find the acute angle between each pair of lines:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 9 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 11 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} -6 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -2 \end{pmatrix}$

Question 30. Find the acute angle between the line $\mathbf{r} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -5 \end{pmatrix}$ and the:

a) x-axis.

b) y-axis.

Question 31. Find the obtuse angle between the following pair of lines: $l_1 : \mathbf{r} = 3\mathbf{i} - 2\mathbf{j} + \lambda(4\mathbf{i} - \mathbf{j})$, $l_2 : \mathbf{r} = 6\mathbf{i} + \mathbf{j} + \mu(3\mathbf{i} + 4\mathbf{j})$.

Question 32. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \end{pmatrix}$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix}$. Find the equation of the line l_2.

Question 33. The line l_1 has equation $\mathbf{r} = 6\mathbf{i} - 3\mathbf{j} + \lambda(2\mathbf{i} - 7\mathbf{j})$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 4 & 1 \\ 2 & -3 \end{pmatrix}$. Find the equation of the line l_2.

Question 34. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} -9 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -8 \end{pmatrix}$. l_1 is mapped to its image l_2 by each of the following transformations. Find the equation of the line l_2 in each case:

a) Stretch, parallel to the x-axis, scale factor 5
b) Enlargement, centred at 0, scale factor 2
c) Reflection in the x-axis
d) Reflection in the y-axis
e) Rotation, anticlockwise about the origin, by 90°
f) Rotation, anticlockwise about the origin, by 180°

Question 35. Find the shortest distance from the origin to the line with equation $y = 2x + 5$.

Question 36. Find the shortest distance from the point $(3, -5)$ to the line with the equation $y = 8x + 1$.

Question 37. Find the shortest distance from the point $(4, -2)$ to the line with equation $\mathbf{r} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Question 38. The shortest distance from the point $(1, k)$ to the line with equation $\mathbf{r} = \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 3 \end{pmatrix}$ is $14\sqrt{34}$ units. Find the integer value of k.

Question 39. Find the shortest distance between the following pair of parallel lines: $l_1 : y = x + 2$, $l_2 : y = x + 8$.

Question 40. Find the shortest distance between the following pair of parallel lines: $l_1 : y = 5x - 20$, $l_2 : y = 5x + 30$.

Question 41. Find the shortest distance between the following pair of parallel lines: $l_1 : y = 2 - 10x$, $l_2 : y = -1 - 10x$.

Question 42. Find the shortest distance between the following pair of parallel lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

Question 43. Find the shortest distance between the following pair of parallel lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 5 \end{pmatrix}.$$

Question 44. It is given that $l_1 : \mathbf{r} = \begin{pmatrix} -4 \\ k \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 8 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ are parallel. It is also given that the shortest distance between l_1 and l_2 is $4\sqrt{29}$ units. Find the possible values of k.

3D Vectors: The Scalar Product

Questions 1–29

Question 1. Find the length of the following vectors:

a) $\mathbf{a} = \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}$

b) $\mathbf{b} = 12\mathbf{j} - 16\mathbf{k}$

Question 2. Given that the scalar product is distributive, find the value of:

a) $\mathbf{i} \cdot \mathbf{j}$

b) $\mathbf{j} \cdot \mathbf{k}$

c) $\mathbf{i} \cdot \mathbf{i}$

d) $5\mathbf{j} \cdot 3\mathbf{j}$

e) $2\mathbf{j} \cdot 6\mathbf{k}$

f) $\mathbf{i} \cdot (\mathbf{i} + \mathbf{j})$

g) $\mathbf{k} \cdot (2\mathbf{i} - \mathbf{j})$

h) $\mathbf{i} \cdot (3\mathbf{i} - 2\mathbf{j} + \mathbf{k})$

i) $(\mathbf{i} + \mathbf{j}) \cdot (\mathbf{i} + \mathbf{k})$

j) $(2\mathbf{i} - 3\mathbf{k}) \cdot (2\mathbf{j} - 3\mathbf{k})$

k) $(\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot (\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$

Question 3. Find the scalar product of:

a) $\mathbf{a} = \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 10 \\ 20 \\ -5 \end{pmatrix}$

b) $\mathbf{a} = 6\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{b} = -3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$

c) $\mathbf{a} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$

d) $\mathbf{a} = 8\mathbf{i} - 15\mathbf{k}$ and $\mathbf{b} = 16\mathbf{i} - 18\mathbf{j}$

Question 4. Find the value of $(2\mathbf{i} + p\mathbf{j} - 3p\mathbf{k}) \cdot (3p\mathbf{i} - 2\mathbf{j} - \mathbf{k})$ in terms of p.

Question 5. It is given that $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}$, and that θ is the acute angle between $\mathbf{a}$ and $\mathbf{b}$. Find the exact value of $\cos\theta$.

Question 6. Use the scalar product to find the acute angle between the following pairs of vectors. Give your answer in degrees to 1 decimal place.

a) $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$

b) $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

c) $\mathbf{a} = \begin{pmatrix} \sqrt{2} \\ 3 \\ 2\sqrt{2} \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 5\sqrt{2} \\ 5 \end{pmatrix}$

d) $\mathbf{a} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$

Question 7. The acute angle between vectors $\mathbf{a}$ and $\mathbf{b}$ is θ. It is given that $\mathbf{a} \cdot \mathbf{b} = 8$, $|\mathbf{a}| = 2$ and $|\mathbf{b}| = \frac{11}{2}$. Find the exact value of $\sin\theta$.

Question 8. The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively relative to an origin O, where $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - \mathbf{j} - \mathbf{k}$. Find the angle OAB.

Question 9. The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively relative to an origin O, where $\mathbf{a} = -4\mathbf{i} + \mathbf{j} - 8\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$. The angle AOB is θ.

a) Find $\cos\theta$.

b) Find the area of the triangle AOB.

Question 10. It is given that $A(-3, -1, 2)$, $B(8, -4, 4)$ and $C(2, -9, -10)$. Find the angle ABC to 1 decimal place.

Question 11. In each case, determine if the following pairs of vectors are perpendicular.

a) $\mathbf{a} = \begin{pmatrix} 6 \\ -1 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ 4 \\ -5 \end{pmatrix}$

b) $\mathbf{a} = -2\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 6\mathbf{j} - 6\mathbf{k}$

c) $\mathbf{a} = \begin{pmatrix} 9 \\ -6 \\ 15 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -5 \\ -10 \\ -1 \end{pmatrix}$

Question 12. It is given that $A(8, -2, 1)$, $B(10, -6, 5)$, $C(10, -4, 2)$ and $D(12, 4, 5)$ are the vertices of a triangular-based pyramid. The angle BAC is θ.

a) Find the exact value of $\sin\theta$.

b) Show that $\overrightarrow{AD}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

c) Given that the volume of a pyramid is $\frac{1}{3}\times$ base area $\times$ perpendicular height, find the volume of the pyramid $ABCD$.

Question 13. It is given that $A(1, 9, -6)$, $B(2, 5, 2)$ and $C(\frac{11}{3}, \frac{28}{3}, \frac{7}{3})$ are the vertices of a triangle. D is the point on AB such that $AD : DB = 2 : 1$.

a) Show that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{DC}$.

b) Find the exact area of the triangle ABC.

Question 14. Determine the value(s) of p that would make the following pairs of vectors perpendicular:

a) $\mathbf{a} = 3p\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - 4p\mathbf{j} + 6\mathbf{k}$

b) $\mathbf{a} = \begin{pmatrix} 16 \\ -4p \\ 7 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 7 \\ 40p \end{pmatrix}$

c) $\mathbf{a} = 5p\mathbf{i} - p\mathbf{j} + \mathbf{k}$ and $\mathbf{b} = p\mathbf{i} - 5\mathbf{j} + 8p\mathbf{k}$

d) $\mathbf{a} = \begin{pmatrix} 4p \\ -6p \\ 12 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3p \\ 2 \\ 6 \end{pmatrix}$

Question 15. The points A, B and C have position vectors $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$ and $\mathbf{c} = 6\mathbf{i} + (p-5)\mathbf{j} - \mathbf{k}$ respectively, where p is a constant.

a) Show that angle ACB can never be $90°$.

b) Find the value of p such that:

(i) angle $ABC = 90°$

(ii) A, B and C are collinear.

Question 16. It is given that $\mathbf{a} = \begin{pmatrix} 3k \\ -2 \\ 5 \end{pmatrix}$, where k is a constant, is perpendicular to $\mathbf{b} = \begin{pmatrix} 4 \\ -3 \\ -2 \end{pmatrix}$. Find the value of k.

Question 17. It is given that $\mathbf{a} = \begin{pmatrix} -6 \\ 2(k+1)^2 \\ 29 \end{pmatrix}$, where k is a constant, is perpendicular to $\mathbf{b} = \begin{pmatrix} 8 \\ -5 \\ 2 \end{pmatrix}$. Find the possible values of k.

Question 18. It is given that $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$. The vector $\mathbf{c} = i\mathbf{i} + m\mathbf{j} + n\mathbf{k}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, where m and n are constants. Find the values of m and n.

Question 19. It is given that $\mathbf{a} = \begin{pmatrix} -13 \\ -4 \\ -9 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ -7 \\ 10 \end{pmatrix}$. The vector $\mathbf{c} = \begin{pmatrix} m \\ n \\ 5 \end{pmatrix}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, where m and n are constants. Find the values of m and n.

Question 20. Find a non-zero vector that is perpendicular to both $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 4\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$.

Question 21. Find a unit vector that is perpendicular to both $\mathbf{a} = -42\mathbf{i} - \mathbf{j} + 6\mathbf{k}$ and $\mathbf{b} = 10\mathbf{i} + 5\mathbf{j} - 5\mathbf{k}$.

Question 22. Find a non-zero vector, in terms of p, that is perpendicular to both $\mathbf{a} = \begin{pmatrix} p \\ 3p \\ 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 4 \\ 3 \end{pmatrix}$.

Question 23. Find, in terms of the constant a, a vector that is perpendicular to both $\mathbf{a} = \begin{pmatrix} 2q \\ -5 \\ 3q \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$.

Question 24. Show that the vectors $\mathbf{a} = \begin{pmatrix} x+1 \\ x-2 \\ 2x+3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 4x \\ 7 \end{pmatrix}$ can never be perpendicular.

Question 25. Show that the vectors $\mathbf{a} = \begin{pmatrix} 3-x \\ 4-3x \\ 5-x \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} x-1 \\ x \\ -5 \end{pmatrix}$ can never be perpendicular.

Question 26. It is given that $\mathbf{a} = \begin{pmatrix} 2 \\ q \\ -3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} q+1 \\ q \\ -1 \end{pmatrix}$, where q is a constant.

a) Find a vector $\mathbf{c}$ that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

b) Given that $\mathbf{d} = \begin{pmatrix} -5q \\ 5 \\ 1 \end{pmatrix}$, show that $\mathbf{c}$ is never perpendicular to $\mathbf{d}$.

Question 27. Find the shortest possible vector of the form $\begin{pmatrix} m \\ n \\ 1 \end{pmatrix}$ that is perpendicular to both $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ x \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -4 \\ 3 \\ x+5 \end{pmatrix}$.

Question 28. Find the shortest possible vector of the form $\begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ that is perpendicular to both $\mathbf{a} = \begin{pmatrix} 1 \\ x \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 1-x \\ 8 \end{pmatrix}$.

Question 29. It is given that $\mathbf{p}$ and $\mathbf{q}$ are nonzero 3D vectors. It is also given that $\mathbf{p}+\mathbf{q}$ is perpendicular to $\mathbf{p}-\mathbf{q}$. Prove that $|\mathbf{p}| = |\mathbf{q}|$.

Equations of Lines in 3D: Vector Equations

Questions 1–59

Question 1. Find the vector equation of the line that:

a) passes through $(2, -7, 1)$ and is parallel to $3\mathbf{i} + 9\mathbf{j} - 4\mathbf{k}$

b) passes through $(3, 8, -3)$ and is parallel to $-4\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$

Question 2. Find a vector equation of the line passing through the following pairs of points:

a) $(0, 0, 0)$ and $(4, -1, 3)$

b) $(1, 4, 1)$ and $(3, 5, 7)$

c) $(2, -4, 5)$ and $(3, -1, 6)$

d) $(9, -8, -2)$ and $(9, -3, -8)$

e) $(6, -4, 13)$ and $(1, -4, 8)$

f) $(-1, 10, 3)$ and $(-9, -4, 3)$

Question 3. It is given that l is the line with equation $\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$. Given also that points A and B lie on the line l with $\lambda = 2$ and $\lambda = -2$ respectively, find the exact length of the line joining A and B.

Question 4. Show that the point $(6, -4, -12)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 6 \\ 27 \end{pmatrix}$.

Question 5. Show that the point $(-12, \frac{31}{2}, -\frac{27}{2})$ does not lie on the line with equation $\mathbf{r} = -2\mathbf{i} + 13\mathbf{j} - 6\mathbf{k} + \lambda(4\mathbf{i} - \mathbf{j} + 7\mathbf{k})$.

Question 6. Determine whether the point $(-2, 16, -40)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 7.5 \\ -3 \\ 2.5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix}$.

Question 7. The point $(5, -10, 2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ -4 \\ k \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$, where k is a constant. Find the value of k.

Question 8. The point $(-3, -83, -98)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 4 \\ 5 \end{pmatrix}$, where k is a constant. Find the value of k.

Question 9. Determine whether the points $A(1, -1, -5)$, $B(-5, -9, -1)$ and $C(16, 19, -15)$ are collinear.

Question 10. Determine whether the points $A(40, -46, -18)$, $B(-8, 18, -10)$ and $C(4, 2, -11)$ are collinear.

Question 11. It is given that $A(7, p, p)$, $B(-1, 11, 27)$ and $C(17, -16, q)$, where p and q are constants, are collinear. Find the value of p and the value of q.

Question 12. Determine whether the following pair of lines are coincident, parallel or neither:

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 8 \\ 13.5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ 8 \\ -9 \end{pmatrix} + \lambda \begin{pmatrix} 0.5 \\ -2 \\ -3.375 \end{pmatrix}.$$

Question 13. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 2 \\ 9 \end{pmatrix}.$$

Question 14. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 2 \\ 10 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 6 \\ -2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 8 \\ -1 \\ 8 \end{pmatrix}.$$

Question 15. Determine whether the following pairs of lines are coincident, parallel, skew or if they intersect. If they intersect, find the coordinates of the point of intersection:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 9 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -9 \\ 16 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} -4 \\ 3 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 6 \\ -\frac{32}{3} \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} -5 \\ -2 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -8 \\ -5 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} -4.5 \\ 11.5 \\ -7.5 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ -9 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ -8 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 6 \\ 7 \end{pmatrix}$

Question 16. It is given that the following lines intersect:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ k \\ 8 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ 5 \\ -9 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -6 \\ 4 \end{pmatrix},$$

where k is a constant.

a) Find the value of k.

b) Find the position vector of the point of intersection of the two lines.

Question 17. It is given that the following lines are perpendicular:

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ -3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 6 \\ k \\ -1 \end{pmatrix},$$

where k is a constant. Find the value of k.

Question 18. It is given that the line l_1 has the equation $\mathbf{r} = \begin{pmatrix} 1 \\ 9 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$, and that the line l_2 is parallel to the vector $24\mathbf{i} + 4\mathbf{j} - 9\mathbf{k}$. Show that l_1 and l_2 are perpendicular.

Question 19. It is given that the line l_1 has the equation $\mathbf{r} = \mathbf{i} + \lambda(2\mathbf{i} - \mathbf{j} - \mathbf{k})$, and that the line l_2 has the equation $\mathbf{r} = 3\mathbf{j} + 4\mathbf{k} + \mu(\mathbf{j} - \mathbf{k})$. Determine whether l_1 and l_2 are perpendicular.

Question 20.

a) Show that $-2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $3\mathbf{i} - 5\mathbf{j} - \mathbf{k}$ are not perpendicular vectors.

b) Find the coordinates of the point of intersection of the two lines $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 6 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} -13 \\ 28 \\ 24 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix}$.

c) It is given that the line l_3 intersects both l_1 and l_2, and is perpendicular to both l_1 and l_2. Find the equation of the line l_3.

Question 21. Find the acute angle between each pair of lines:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -7 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} -6 \\ 3 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ -10 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ -8 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ 4 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} -7 \\ 6 \\ 6 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -9 \\ 2 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} -11 \\ 8 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} -15 \\ 3 \\ 1 \end{pmatrix}$

Question 22. Find the acute angle between the line $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and the:

a) x-axis
b) y-axis
c) z-axis

Question 23. Find the obtuse angle between the following pair of lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -12 \\ 3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 1 \\ 5 \end{pmatrix}.$$

Question 24. Find the obtuse angle between the following pair of lines:

$$l_1 : \mathbf{r} = 2\mathbf{i} + 5\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + \mathbf{k}), \quad l_2 : \mathbf{r} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k} + \mu(2\mathbf{i} + 4\mathbf{j} + 11\mathbf{k}).$$

Question 25. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 6 \\ 2 \end{pmatrix}$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} -1 & 2 & 3 \\ 1 & -1 & 2 \\ 0 & 2 & 1 \end{pmatrix}$. Find the equation of the line l_2.

Question 26. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 7 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix}$. l_1 is mapped to its image l_2 by each of the following transformations. Find the equation of the line l_2 in each case:

a) Stretch, parallel to the x-axis, scale factor 2
b) Enlargement, centred at O, scale factor 3
c) Reflection in the plane $x = 0$
d) Reflection in the plane $y = 0$
e) Rotation, anticlockwise about the y-axis, by 90°
f) Rotation, anticlockwise about the z-axis, by 90°

Question 27. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$. The line l_1 is mapped to its image l_2 by the transformation associated with each of the following matrices. Find the acute angle between l_1 and l_2 in each case:

a) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix}$

b) $\begin{pmatrix} -2 & 3 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 5 & -3 & -4 \\ 2 & 8 & -3 \\ -5 & -4 & -1 \end{pmatrix}$

Question 28. Write down the coordinates of a general point, in terms of the given parameter, that lies on each of the following lines:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} 5 \\ -9 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ -3 \\ 16 \end{pmatrix}$

Question 29. The line l has equation $\mathbf{r} = \begin{pmatrix} -4 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix}$. State the coordinates of the point on the line with the given value of λ:

a) $\lambda = 1$
b) $\lambda = -1$
c) $\lambda = -4$

Question 30. It is given that the point A has coordinates $(2, 1, -5)$. The point B lies on the line l which has the equation $\mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ -7 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$. Find $\overrightarrow{AB}$ in terms of λ.

Question 31. It is given that the point P has coordinates $(3, -7, 2)$. The point Q lies on the line l which has the equation $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -2 \\ 5 \end{pmatrix}$. Find $\overrightarrow{QP}$ in terms of λ.

Question 32. Find the value of λ in each case such that the vector $\begin{pmatrix} 2-\lambda \\ 1+5\lambda \\ 2\lambda - 1 \end{pmatrix}$ is perpendicular to:

a) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$

c) $\begin{pmatrix} -1 \\ 3 \\ -5 \end{pmatrix}$

Question 33. The line l_1 passes through the points A and B with coordinates $(3, 7, -1)$ and $(2, 5, -3)$ respectively. The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 6 \\ -2 \\ -3 \end{pmatrix}$.

a) Find a vector equation of the line l_1.

b) Show that $P(38, -17, -15)$ lies on l_2. It is given that C lies on l_1 such that CP is perpendicular to l_2.

c) Find the coordinates of C.

Question 34. The points A and B have coordinates $(2, -1, 2)$ and $(3, 8, -2)$ respectively. The point C is such that $\overrightarrow{OC} = 99\overrightarrow{OA}$. The line l is parallel to AB and passes through C.

a) Find a vector equation of the line l. The point D lies on l such that angle $ABD = 90°$.

b) Find the coordinates of D.

Question 35. Two lines are given by

$$l_1 : \mathbf{r} = \begin{pmatrix} 10.25 \\ -6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} a \\ 1 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ b \end{pmatrix},$$

where a and b are positive constants. It is also given that the lines l_1 and l_2 intersect and form an angle of $120°$. Find the exact values of the constants a and b.

Question 36. The points A and B have position vectors $2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $3\mathbf{i} - \mathbf{j} - 4\mathbf{k}$ relative to a fixed origin O. The points A and B lie on the line l.

a) Find a vector equation for the line l.

The point C has position vector $a\mathbf{i} + \mathbf{j} - \mathbf{k}$, where a is a constant. It is given that AC is perpendicular to the line l.

a) Find the value of a.

b) Find the area of the triangle ACB.

Question 37. It is given that the line l has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -5 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix}$.

a) The point A lies on the line l when $\lambda = -2$. Find the coordinates of A.

b) Write down the vector equation of a line that is perpendicular to l and passes through A.

Question 38. It is given that:

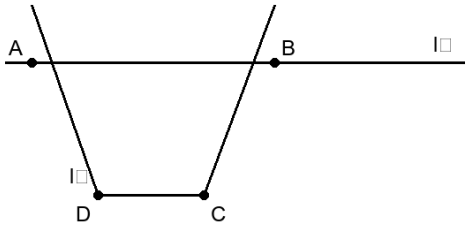
$$l_1 : \mathbf{r} = \begin{pmatrix} -7 \\ 3 \\ 10 \end{pmatrix} + \alpha \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix},$$

$$l_3 : \mathbf{r} = \begin{pmatrix} -21 \\ 5 \\ 23 \end{pmatrix} + \gamma \begin{pmatrix} 4 \\ -1 \\ -5 \end{pmatrix}, \quad l_4 : \mathbf{r} = \begin{pmatrix} 19 \\ 1 \\ -9 \end{pmatrix} + \delta \begin{pmatrix} -6 \\ 0 \\ 3 \end{pmatrix}.$$

a) Show that the points where these four lines intersect form a parallelogram but not a rhombus.

b) Hence find the exact area of the parallelogram.

Question 39. The diagram below shows the symmetrical trapezium $ABCD$, where $|AD| = |BC|$.



The points A and B have coordinates $(6, -6, 1)$ and $(14, -12, 5)$ respectively. A and B lie on the line l_1.

a) Find a vector equation of the line l_1.

It is given that the line l_2 passes through A and D and has equation $\mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$.

a) Show that l_1 and l_2 intersect at A.

b) D lies on l_2 when $\mu = 4$. Find the coordinates of D.

c) Find the coordinates of C.

Question 40. It is given that A, B, C and D have coordinates $(4, 7, -6)$, $(1, 4, -6)$, $(3, 2, -9)$ and $(3, 3, -9)$ respectively. A and B lie on the line l_1, while C and D lie on the line l_2. l_1 and l_2 intersect at the point P.

a) Find the coordinates of P.

b) E lies on l_2 such that $|BP| = |PE|$. Find the possible coordinates of E.

Question 41. A, B, C and D have coordinates $(4, -5, 4-\sqrt{6})$, $(4+\sqrt{6}, -5-3\sqrt{6}, -2-\sqrt{6})$, $(3+\sqrt{6}, 6, 1)$ and $(\sqrt{6}-9, 1-6\sqrt{6}, 5-2\sqrt{6})$ respectively. A and B lie on the line l_1, while C and D lie on the line l_2. A and C lie on a sphere, whose centre is the point of intersection of l_1 and l_2. Find the volume of the sphere.

Question 42. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -3 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 5 \\ -8 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 16 \\ -8 \\ -23 \end{pmatrix}.$$

The lines l_1 and l_2 intersect at the point P.

a) Find the coordinates of P. The point A lies on l_1 such that $|AP| = 849\sqrt{43}$. The point B lies on l_2 such that angle $ABP = 90°$.

b) Find the two possible coordinates of B.

Question 43. A, B and C have coordinates $(3, 4, -5)$, $(-1, 3, -2)$ and $(3, p, q)$ respectively, where p and q are constants and $p > 0$. A and B lie on the line l.

a) Find the coordinates of C, given that C is $\sqrt{90}$ units from A and that angle $CAB = 90°$. The point D lies on the line l such that the triangle CAD has twice the area of the triangle CAB.

b) Find the possible coordinates of D.

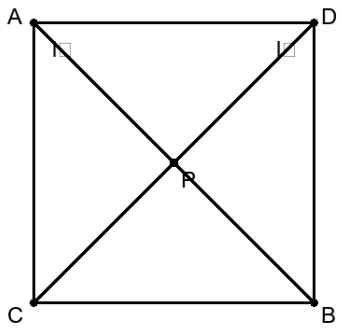
Question 44. It is given that the line l_1 passes through the points A and B with coordinates $(-1 + 3\sqrt{89}, 5 - 2\sqrt{89}, 9 + 6\sqrt{89})$ and $(-1 - 3\sqrt{89}, 5 + 2\sqrt{89}, 9 - 6\sqrt{89})$ respectively.

a) State the midpoint of A and B.

The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ 23 \\ 13 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 9 \\ 2 \end{pmatrix}$.

a) l_1 and l_2 intersect at the point P. Find the coordinates of P.

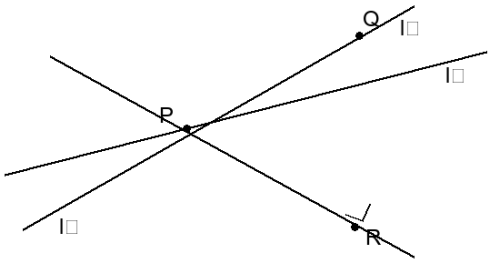
b) The points C and D lie on l_2 such that $ADBC$ is a square, as shown in the diagram.



Find the coordinates of C and D.

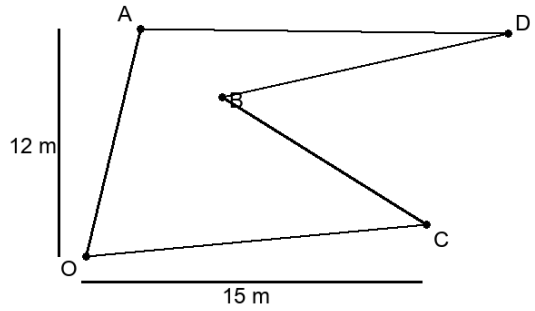
Question 45. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ -3 \\ -9 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -6 \\ 1 \\ 2 \end{pmatrix}.$$



The point P has coordinates $(18, 4, -1)$ and is the point where l_2 and l_3 intersect. The line l_3 is parallel to l_1. Q lies on the line l_3. R lies on the line l_2 when $\mu = -27$. It is given that QR is perpendicular to l_2. Find the coordinates of Q.

Question 46. A drone light show involves two drones that are programmed to follow straight paths at different angles. The drones are programmed to remain within the symmetrical trapezium shown in the diagram, as seen from above.



Take the point O to have coordinates $(0, 0, 0)$. Drone 1 starts from 1 metre above point A and follows a straight path to reach a height of 15 metres above point C. Meanwhile, Drone 2 starts from 4 metres above point B and follows a straight path to reach a height of k metres above point D.

a) Given that the drones' flight paths intersect one another, find the value of k.

It is given that the drones are programmed to set off at the same time and both fly at constant speeds. It takes Drone 1 five seconds to fly from its initial position 1 metre above A to its final position 15 metres above C.

a) How fast would Drone 2 need to fly to crash into Drone 1?

b) Describe two assumptions you have made in answering part (b).

Question 47. The points A and B have coordinates $(6, 2, -2)$ and $(9, 3, -3)$ respectively.

a) Show that the vector equation of the line passing through A and B can be given by $\mathbf{r} = \lambda \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$.

The points A and B are transformed to their images C and D respectively by a rotation, anticlockwise about the z-axis by $90°$.

a) Find the equation of the line passing through C and D.

b) Find the acute angle between the line in part (a) and the line found in part (b).

Question 48. It is given that the line l_1 has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 1 & 2 & 0 \\ -1 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Find the equation of the line l_2.

Question 49. It is given that the line l_1 has equation $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k} + \lambda(3\mathbf{i} - \mathbf{j} - \mathbf{k})$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix}$. Find the equation of the line l_2.

Question 50. It is given that $l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 6 \\ -11 \end{pmatrix}$. Show that the line l_1 can be mapped to the line l_2 by the transformation represented by the matrix $\begin{pmatrix} 1 & -2 & 1 \\ 1.5 & -1 & 0.25 \\ 1 & 0.5 & -3 \end{pmatrix}$.

Question 51. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 2 \\ -5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 26 \\ -15 \\ -10 \end{pmatrix} + \mu \begin{pmatrix} a \\ 16 \\ b \end{pmatrix}, \quad A = \begin{pmatrix} p & 1 & -3.5 \\ 3 & -3 & 2 \\ 2.5 & -1 & q \end{pmatrix}.$$

Given also that $\det A \neq 0$, l_1 can be mapped to l_2 by the transformation represented by matrix A. Find the values of the constants a, b, p and q.

Question 52. It is given that the line l has equation $\mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$. l intersects a circle centred at $(1, 1, 0)$ with radius $\sqrt{14}$ at two distinct points. Find the coordinates of these two points.

Question 53. The plane Π has equation $3x - 2y + 5z = 6$. The line l has equation $\mathbf{r} = \begin{pmatrix} 40 \\ 7 \\ -20 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 9 \\ 6 \end{pmatrix}$. Show that l lies in the plane Π.

Question 54. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 9 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ -1 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ -8 \end{pmatrix}.$$

Find a vector equation of the plane Π.

Question 55. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ -6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 6 \\ -3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 3 \\ 25 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 9 \\ 18 \end{pmatrix}.$$

Find a cartesian equation for the plane Π.

Question 56. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 9 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ k \\ -5 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 5 \\ 6 \end{pmatrix}.$$

Find the value of the constant k.

Question 57. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 3 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix},$$

where k is a constant. Find a cartesian equation of the plane Π that is independent of k.

Question 58. Write the following vector equations of lines in cartesian form:

a) $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \\ -7 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$

c) $\mathbf{r} = \begin{pmatrix} 4 \\ -8 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 6 \end{pmatrix}$

d) $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -8 \end{pmatrix}$

e) $\mathbf{r} = \begin{pmatrix} 7 \\ -9 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -5 \\ 0 \end{pmatrix}$

f) $\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Question 59. Write the following vector equations of lines in vector product form:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} 0 \\ -6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -2 \\ -5 \end{pmatrix}$

Equations of Lines in 3D: Cartesian Equations

Questions 1–20

Question 1. Find the cartesian equation of the line that:

a) passes through $(4, 1, -2)$ and is parallel to $2\mathbf{i} + 3\mathbf{j} + 8\mathbf{k}$
b) passes through $(2, -8, -5)$ and is parallel to $9\mathbf{i} - \mathbf{j} - 2\mathbf{k}$
c) passes through $(-3, -5, 6)$ and is parallel to $2\mathbf{i} - \mathbf{j} - 5\mathbf{k}$
d) passes through $(1, -1, 0)$ and is parallel to $-\mathbf{i} + 3\mathbf{j} + 10\mathbf{k}$
e) passes through $(0, 8, -3)$ and is parallel to $-\mathbf{i} - 5\mathbf{j} + 9\mathbf{k}$
f) passes through $\left(\frac{1}{2}, -3, -\frac{1}{3}\right)$ and is parallel to $5\mathbf{i} - 2\mathbf{j} - 10\mathbf{k}$
g) passes through $\left(-\frac{1}{4}, \frac{1}{2}, \frac{2}{3}\right)$ and is parallel to $10\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$
h) passes through $(2, -5, -7)$ and is parallel to $3\mathbf{i} + 4\mathbf{j}$
i) passes through $(-7, 3, 12)$ and is parallel to $8\mathbf{i} - 3\mathbf{k}$
j) passes through $(-4, 3, 3)$ and is parallel to $-5\mathbf{j} + 6\mathbf{k}$
k) passes through $(-8, 3, -7)$ and is parallel to $\mathbf{i}$
l) passes through $\left(\frac{3}{4}, -\frac{5}{3}, \frac{9}{4}\right)$ and is parallel to $-16\mathbf{i} + 3\mathbf{k}$

Question 2. Find the coordinates of the point when $z = 6$ on the line with equation

$$\frac{x-1}{3} = \frac{y+2}{4} = \frac{z-5}{2}.$$

Question 3. Find the coordinates of the point when $y = 8$ on the line with equation

$$\frac{x+5}{6} = \frac{2-y}{10} = \frac{z+8}{-3}.$$

Question 4. Show that the point $(-5, 20, 6)$ lies on the line with equation

$$\frac{x-3}{-2} = \frac{y-4}{4} = \frac{z+6}{3}.$$

Question 5. Determine whether the point $(2, -8, 3)$ lies on the line with equation

$$\frac{x-9}{3} = \frac{y-13}{9}, \quad z = 3.$$

Question 6. Determine whether the point $(15, 11, -10)$ lies on the line with equation

$$\frac{x+2}{8} = \frac{y+57}{32} = \frac{z+18}{4}.$$

Question 7. Find a cartesian equation of the line passing through the following pairs of points:

a) $(0, 0, 0)$ and $(2, 3, 6)$
b) $(1, 2, -4)$ and $(3, 5, -7)$
c) $(4, 0, -8)$ and $(5, 1, 0)$
d) $(2, 5, -1)$ and $(2, -3, 2)$
e) $(3, 8, 1)$ and $(9, 8, -6)$
f) $(5, -3, -2)$ and $(6, -7, -2)$

Question 8. Determine whether the following pair of lines are parallel or not:

$$l_1 : \frac{2x-3}{6} = \frac{5y+11}{-4} = \frac{8z-9}{12}, \qquad l_2 : \frac{1-3x}{9} = \frac{10y-13}{8} = \frac{5-2z}{3}.$$

Question 9. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \frac{x-3}{2} = \frac{y+2}{3} = \frac{z-74}{10}, \qquad l_2 : \frac{x+1}{3} = \frac{y-4}{6} = \frac{z+2}{8}.$$

Question 10. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \frac{5x+25}{-8} = \frac{5y-25}{2} = \frac{5z+3}{-4}, \qquad l_2 : \frac{5x+20}{-3} = \frac{5y+25}{-3} = \frac{5z-45}{2}.$$

Question 11. Determine whether the following pair of lines are coincident, parallel, skew or if they intersect. If they intersect, find the coordinates of the point of intersection.

a) $l_1 : \frac{x+1}{4} = \frac{y-3}{-5} = \frac{2z-7}{-4}$, $l_2 : \frac{x-9}{-4} = \frac{y+2}{5} = \frac{3z+1}{6}$

b) $l_1 : \frac{x+2}{3} = \frac{4y-1}{3} = 2 - z$, $l_2 : \frac{x+3}{3} = \frac{8y+3}{6} = 3 + z$

c) $l_1 : \frac{4x-1}{2} = y + 7 = \frac{3-z}{8}$, $l_2 : \frac{6x-7}{3} = \frac{2y+1}{4} = \frac{2z-3}{2}$

d) $l_1 : \frac{1-4x}{8} = \frac{y+5}{3} = \frac{5z-65}{4}$, $l_2 : \frac{3x-2}{5} = \frac{7-4y}{9} = 3 - z$

Question 12. It is given that the following lines intersect:

$$l_1 : \frac{x-6}{-3} = \frac{y+1}{2} = \frac{z-7}{4}, \qquad l_2 : \frac{x-15}{k} = \frac{y+3}{9} = \frac{z+12}{6},$$

where k is a constant.

a) Find the value of k.

b) Find the position vector of the point of intersection of the two lines.

Question 13. Find the acute angle between each pair of lines:

a) $l_1 : \frac{x-3}{2} = \frac{y+7}{9} = \frac{z-3}{4}, \quad l_2 : \frac{x+7}{4} = \frac{y-1}{2} = \frac{z+8}{3}$

b) $l_1 : \frac{x+2}{-8} = \frac{y-1}{2} = z + 2, \quad l_2 : \frac{x-3}{4} = \frac{y+2}{3} = \frac{z-4}{2}$

c) $l_1 : \frac{3-x}{2} = \frac{y+1}{-3} = 3 - z, \quad l_2 : x + 2 = 3 - y = 4 - z$

d) $l_1 : x = 3, \quad \frac{2-y}{2} = \frac{z+8}{1}, \quad l_2 : x = 8, \quad y - 9 = \frac{z-4}{5}$

e) $l_1 : \frac{2x-3}{4} = \frac{7-3y}{6} = \frac{2z+1}{4}, \quad l_2 : \frac{2x+1}{3} = \frac{6-4y}{5} = 10 - z$

f) $l_1 : 5 - x = \frac{y-9}{4}, \quad z = 5, \quad l_2 : \frac{8x+3}{3} = \frac{4-z}{5}, \quad y = -2$

Question 14. Two small intergalactic asteroids are travelling on different straight-line paths, which can be modelled by the equations:

$$l_1 : \frac{x - 2140}{10} = \frac{y - 425}{9}, \quad z = 670, \qquad l_2 : \frac{x + 200}{15k} = \frac{y - 380}{18} = \frac{475 - z}{5},$$

where k is a constant. The two lines intersect.

a) Find the value of k.

b) Find the coordinates of where the two lines intersect.

c) It is given that at time T seconds, the asteroids are both the same distance from where their paths intersect. Explain why it is not necessarily the case that the asteroids will impact one another.

Question 15. The line l_1 has the equation:

$$\frac{x+1}{4} = 3 - y = \frac{3z+4}{6}$$

and l_2 passes through $(5, 0, 4)$ and $(6, k, k+1)$, where k is a constant. Given that l_1 and l_2 are perpendicular,

a) find the value of k

It is given that l_3 passes through $(5, 0, 4)$ and $(6, k, k-1)$ and that θ is the angle between l_1 and l_3.

a) Show that $\cos\theta = -\frac{2\sqrt{6}}{21}$.

Question 16. Two lines are given by

$$l_1 : \frac{5-x}{2} = 3 - y = 2z, \qquad l_2 : \frac{x-2}{8-k} = \frac{y+5}{2p+5} = \frac{z-6}{-4}.$$

Find the values of k and p given that:

a) l_1 and l_2 are parallel

b) l_1 and l_2 intersect and are perpendicular

Question 17. The plane Π contains the following two lines:

$$l_1 : \frac{x-3}{7} = \frac{2y+1}{2} = 1-z, \qquad l_2 : \frac{x-8}{3} = \frac{2y-21}{6} = \frac{2z+8}{-3}.$$

Find a vector equation of the plane Π.

Question 18. The plane Π contains the following two lines:

$$l_1 : 3-x = \frac{y-4}{-5} = z+1, \qquad l_2 : \frac{x+2}{2} = 2-y = \frac{z+20}{6}.$$

Find a cartesian equation of the plane Π.

Question 19. Write the following cartesian equations of lines in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$:

a) $\frac{x+1}{2} = \frac{y-1}{3} = \frac{z+2}{-3}$

b) $\frac{x}{8} = \frac{y+9}{4} = \frac{z-1}{-5}$

c) $\frac{x+8}{3} = \frac{2-y}{3} = \frac{z}{10}$

d) $\frac{7-x}{3} = \frac{y-8}{12} = z + 3$

e) $\frac{2x-3}{5} = \frac{3y+1}{2} = \frac{5z-3}{2}$

f) $\frac{3-4x}{6} = \frac{5-2y}{2} = \frac{6-8z}{-5}$

g) $x = -4, \quad \frac{y-3}{-4} = \frac{z+12}{7}$

h) $x - 9 = \frac{2y-4}{-5}, \quad z = 1$

i) $x = 6 - z, \quad y = 5$

Question 20. Write the following cartesian equations of lines in vector product form:

a) $\frac{x-3}{5} = \frac{y-2}{2} = \frac{z-4}{6}$

b) $\frac{x}{9} = \frac{2-y}{3} = 5 - z$

c) $3 - x = \frac{y+1}{2} = 3 - z$

d) $\frac{3x+2}{5} = 6 - y, \quad z = 2$

e) $x = \frac{1}{3}, \quad \frac{4y-3}{2} = \frac{5-3z}{5}$

f) $x = 4, \quad 3y = 4 - 2z$

Equations of Planes: Cartesian Equations

Questions 1–36

Question 1. It is given that the plane Π has equation $2x + 8y - 3z = 4$. Show that:

a) $(3, 1, \frac{10}{3})$ lies on the plane Π

b) $(21, -2, 7)$ does not lie on the plane Π

Question 2. The point A has coordinates $(3, -5, 4)$. Determine which of the following planes contains the point A:

a) $\Pi_1 : 2x + y + 6z = 24$

b) $\Pi_2 : -5x - 2y + 3z = 7$

c) $\Pi_3 : 8x + 3y - 8z = -23$

d) $\Pi_4 : \frac{1}{2}x + \frac{2}{3}y + 3z = 6$

Question 3. It is given that the plane Π has equation $3x - 5y + z = 9$. Show that:

a) contains the vector $\begin{pmatrix} -16 \\ -7 \\ 13 \end{pmatrix}$

b) contains the vector $\begin{pmatrix} 13 \\ 4 \\ -19 \end{pmatrix}$

c) does not contain the vector $\begin{pmatrix} -13 \\ -7 \\ -16 \end{pmatrix}$

Question 4. Write down a cartesian equation of the plane passing through the given point and perpendicular to the given vector:

a) $A(1, 4, 1)$, $\mathbf{n} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$

b) $A(6, 1, -2)$, $\mathbf{n} = \begin{pmatrix} -1 \\ 4 \\ -3 \end{pmatrix}$

c) $A(-3, -4, -9)$, $\mathbf{n} = \begin{pmatrix} 5 \\ -1 \\ -2 \end{pmatrix}$

d) $A(\frac{1}{2}, -\frac{3}{2}, \frac{5}{2})$, $\mathbf{n} = \begin{pmatrix} -2 \\ 3 \\ 10 \end{pmatrix}$

e) $A(3, 0, -2)$, $\mathbf{n} = \begin{pmatrix} 0 \\ 7 \\ -8 \end{pmatrix}$

f) $A(-\frac{1}{3}, \frac{1}{4}, \frac{2}{3})$, $\mathbf{n} = \begin{pmatrix} -4 \\ 15 \\ 0 \end{pmatrix}$

Question 5. The point $A(2,1,3)$ lies on a plane. The vector $\mathbf{n} = \begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix}$ is perpendicular to the plane.

a) Find the cartesian equation of the plane.
b) Determine whether the points $B(3,1,-5)$ and $C(-2,0,8)$ lie on the plane.

Question 6. Find the cartesian equation of the plane that is parallel to $4x - y + z + 2 = 0$ and contains the point $(1,1,4)$.

Question 7. Find the cartesian equation of the plane that is parallel to $-3x+8y-z-7=0$ and contains the point $(2, \frac{1}{2}, -8)$.

Question 8. Find the cartesian equation of the plane that is parallel to $kx+3y-2kz-1=0$ and contains the point $(3, -4k, 5)$, where k is a constant.

Question 9. It is given that the plane Π with equation $-x + 6y - 2z = 3$ contains the vector $\mathbf{i} + \mathbf{j} + p\mathbf{k}$.

a) Find the value of p.

b) Hence find the cartesian equation of a plane that is perpendicular to Π and which passes through the point $(2, 1, 1)$.

Question 10. The plane Π_1 has equation $kx - 6y + z = 7$, where k is a constant, and the plane Π_2 has equation $4x + 8y - 3z = -2$. It is given that the angle between the two planes Π_1 and Π_2 is $90°$. Find the value of k.

Question 11. The plane Π_1 has equation $5x + y - kz = 3$ and the plane Π_2 has equation $2kx - 7y + 2z = 9$, where k is a constant. Given that Π_1 and Π_2 are perpendicular, find the value of k.

Question 12. The plane Π_1 has equation $(k+1)x-2y+3kz = 8$ and the plane Π_2 has equation $4kx-(k+2)y-10z = 1$, where k is a constant. Given that Π_1 and Π_2 are perpendicular, find the possible values of k.

Question 13. Find the acute angle, to 1 decimal place, between:

a) the vector $\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ and the plane $x + 4y + 3z = 1$

b) the vector $-\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}$ and the plane $2x - y + 2z = -5$

c) the vector $\mathbf{i} + 8\mathbf{j}$ and the plane $5x - 4z = 2$

d) the vector $9\mathbf{i} - 3\mathbf{k}$ and the plane $-x + 3y + 2z = 25$

Question 14. It is given that the plane Π has equation $x - y - 2z = 3$. Find the acute angle the plane makes with:

a) the XY plane

b) the XZ plane

c) the YZ plane

Question 15. A new road is to be built that goes up a hill. It can be modelled as a plane with equation $2x + 3y - 8z = 4$. The new road connects to a street that can be modelled as a plane with equation $z = 2$. If the angle between the two roads is greater than a 20% incline, then the new road is deemed too steep and is rejected. Determine if the new road will be rejected.

Question 16. Find the acute angle between the following pairs of planes:

a) $\Pi_1 : x + y - z = 2, \quad \Pi_2 : x + y - 2z = 3$

b) $\Pi_1 : 3x + 2y + z = 4, \quad \Pi_2 : 8x + y + 2z = 9$

c) $\Pi_1 : 4x - y - z = 8, \quad \Pi_2 : -x + y + z = -4$

d) $\Pi_1 : -5x - 2y + z + 3 = 0, \quad \Pi_2 : 9x + 10y - 6z - 4 = 0$

e) $\Pi_1 : x - 4z = 9, \quad \Pi_2 : y + 11z = 1$

f) $\Pi_1 : \sqrt{2}x + \sqrt{6}y - z = -10, \quad \Pi_2 : 2\sqrt{2}x + 2\sqrt{3}y - z = 7$

Question 17. Given the equations of the planes $\Pi_1 : x + 12y = 1$ and $\Pi_2 : x + 6y + kz = 2$, where k is a constant, find the range of values of k so that the acute angle between Π_1 and Π_2 is less than $5°$.

Question 18. Given the equations of the planes $\Pi_1 : kx + y + 2z = 3$ and $\Pi_2 : 20y + 39z = 50$, where k is a constant, find the range of values of k so that the acute angle between Π_1 and Π_2 is greater than $1°$.

Question 19. It is given that the plane Π_1 has equation $6x - y + kz = 8$, where k is a positive constant, and that the plane Π_2 has equation $3x - 3y = 5$. It is also given that the angle between the two planes Π_1 and Π_2 is $45°$. Find the exact value of k.

Question 20. It is given that the plane Π_1 has equation $9x + ky - 3z = 2$, where k is a positive constant, and that the plane Π_2 has equation $2x - 2z = -5$. It is also given that the angle between the two planes Π_1 and Π_2 is $30°$. Find the exact value of k.

Question 21. It is given that the plane Π_1 has equation $8x + y + 2kz = -20$, where k is a negative constant, and that the plane Π_2 has equation $4x - 2y = 15$. It is also given that the angle between the two planes Π_1 and Π_2 is $60°$. Find the exact value of k.

Question 22. It is given that the plane Π_1 has equation $8x-4y+z = 5$ and the plane Π_2 has equation $4x+ky-2z = 2$, where k is a constant. It is also given that the acute angle θ between the planes Π_1 and Π_2 is such that $\cos\theta = \frac{1}{9}$. Show that $k = 8 - \frac{4}{\sqrt{3}}$.

Question 23. It is given that the plane Π_1 has normal vector $\mathbf{n}_1 = 10\mathbf{i} - 5\mathbf{j} + p\mathbf{k}$, where p is a negative constant, and that the plane Π_2 has normal vector $\mathbf{n}_2 = 4\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}$. It is also given that:

- the point $(2, 1, -5)$ lies on Π_1
- the angle between the two planes Π_1 and Π_2 is $60°$

Find the cartesian equation of the plane Π_1.

Question 24. It is given that the vector $\mathbf{m} = p\mathbf{i} + q\mathbf{j} + r\mathbf{k}$, where constants $p, q > 0$ and $r < 0$, lies in the plane $3x + z = 3$ and that $|\mathbf{m}| = 4$. It is also given that the angle between $\mathbf{m}$ and the vector $\mathbf{n} = \mathbf{j}$ is $45°$. Find the vector $\mathbf{m}$.

Question 25. It is given that the vector $\mathbf{p} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, where a, b and c are positive constants, lies in the plane $12x - 4y = -3$ and that $|\mathbf{p}| = 6\sqrt{10}$. It is also given that the angle between $\mathbf{p}$ and the vector $\mathbf{q} = \mathbf{k}$ is $30°$. Find the vector $\mathbf{p}$.

Question 26. It is given that the vector $\mathbf{v}$ is a unit vector that is parallel to the plane $x - 2y = 5$. It is also given that $\mathbf{v}$ makes an angle $45°$ with the normal to the plane $8x - 3y + 5z = 1$. Find the two possible vectors $\mathbf{v}$.

Question 27. The points $A(2, 3, -5)$, $B(9, 1, 0)$ and $C(-7, 3, 2)$ all lie in the plane Π.

a) Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

b) Show that the vector $7\mathbf{i} + 47\mathbf{j} + 9\mathbf{k}$ is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

c) Hence find the equation of the plane Π in the form $ax + by + cz = d$, where a, b, c and d are integers.

Question 28. The points $P(-1, 2, 6)$, $Q(3, 2, 8)$ and $R(4, -3, -5)$ all lie in the plane Π.

a) Find $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

b) Show that the vector $5\mathbf{i} + 27\mathbf{j} - 10\mathbf{k}$ is perpendicular to both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

c) Hence find the equation of the plane Π in the form $ax + by + cz = d$, where a, b, c and d are integers.

Question 29. The points $A(4, 0, -2)$, $B(6, -1, 0)$ and $C(2, 1, -3)$ all lie in the plane Π.

a) Find $\overrightarrow{AB} \times \overrightarrow{AC}$.

b) Find the equation of the plane Π in the form $ax + by + cz = d$, where a, b, c and d are integers.

Question 30. The plane with cartesian equation $ax + by + cz = d$, where a, b, c and d are constants, contains the points $(2, 1, 1)$, $(0, 1, 3)$ and $(1, 0, 4)$. Find the values of a, b, c and d.

Question 31. Find the cartesian equation of the plane that contains the three points:

a) $(1, 0, 1)$, $(1, 1, 1)$ and $(0, 1, 0)$

b) $(2, 1, 1)$, $(-1, 1, 2)$ and $(3, -1, -1)$

c) $(-1, 4, 5)$, $(3, 2, -5)$ and $(4, 2, -3)$

d) $(10, -20, 30)$, $(-20, 40, 50)$ and $(0, 0, -10)$

Question 32. It is given that $A(3, 2, -5)$, $B(4, 1, -2)$, $C(2, 0, 6)$ and D are coplanar. Find the value of k in each of the following cases, where the coordinates of D are:

a) $(k, 1, 2)$

b) $(3, k, 2)$

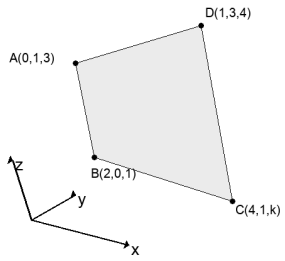
c) $(0, 5, k)$

Question 33. It is given that

$$A = \begin{pmatrix} 2 & -3 & k \\ 4 & 1 & -10 \\ -2 & -11 & 5 \end{pmatrix},$$

where k is a constant. For one particular value of k, the linear transformation associated with the matrix A maps all points to the plane with equation $3x - 2y - z = 0$. Find this value of k.

Question 34. It is given that $A(0, 1, 3)$, $B(2, 0, 1)$, $C(4, 1, k)$ and $D(1, 3, 4)$ all lie on a plane, as shown in the diagram below, where k is a constant.



a) Find the equation of the plane in the form $ax + by + cz = d$, where a, b, c and d are integers.
b) Find the value of the constant k.
c) Find the acute angle the plane makes with the plane with equation $z = 0$.

Question 35. It is given that $A(1,2,-1)$, $B(2,0,-2)$, $C(0,1,1)$ and $D(3,-2,1)$. Additionally, the plane Π_1 contains A, B and C and the plane Π_2 contains A, B and D. Find the acute angle between Π_1 and Π_2.

Question 36. Points $A(0,0,\sqrt{5})$, $B(\frac{\sqrt{15}}{2},0,-\frac{\sqrt{5}}{2})$, $C(-\frac{\sqrt{15}}{2},0,-\frac{\sqrt{5}}{2})$ and $D(0,\sqrt{10},0)$ form a regular tetrahedron.

a) Show that the plane containing A, B and D can be written as $3\sqrt{2}x+\sqrt{3}y+\sqrt{6}z=\sqrt{30}$.

b) Show that the plane containing A, C and D can be written as $3\sqrt{2}x-\sqrt{3}y-\sqrt{6}z=-\sqrt{30}$.

c) Find the acute angle between the ABD and ACD faces of the tetrahedron.

Equations of Planes: Scalar Product Equations

Questions 1–11

Question 1. Write down the equation of the plane passing through the point $(-1, 8, 2)$ and perpendicular to the vector $-2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, giving your answer in the form $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$.

Question 2. It is given that the planes Π_1 and Π_2 have equations $(\mathbf{r}-\mathbf{i})\cdot(2\mathbf{i}+\mathbf{k}) = 0$ and $(\mathbf{r}-3\mathbf{i}+2\mathbf{j})\cdot(6\mathbf{i}-\mathbf{j}+2\mathbf{k}) = 0$ respectively. Find the acute angle between Π_1 and Π_2, giving your answer in degrees to 1 decimal place.

Question 3. It is given that the plane Π has equation

$$\left(\mathbf{r} - \begin{pmatrix} 5 \\ 2 \\ -3 \end{pmatrix}\right) \cdot \begin{pmatrix} -6 \\ 5 \\ -1 \end{pmatrix} = 0.$$

Show that the point with coordinates $(3, \frac{8}{5}, 7)$ lies on the plane.

Question 4. It is given that the plane Π has equation

$$\mathbf{r} \cdot \begin{pmatrix} -9 \\ 8 \\ -16 \end{pmatrix} = 6.$$

Show that the point $(2, -5, -4)$ lies on the plane Π.

Question 5. Write the equation of each of these planes in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers:

a) $\mathbf{r} \cdot \mathbf{i} = 2$
b) $\mathbf{r} \cdot \mathbf{j} = 0$
c) $\mathbf{r} \cdot \mathbf{k} = -3$
d) $\mathbf{r} \cdot (3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) = 9$
e) $\mathbf{r} \cdot (10\mathbf{i} - 50\mathbf{j} + 25\mathbf{k}) = 30$

Question 6. It is given that the plane Π is perpendicular to the vector $2\mathbf{i} - 3\mathbf{j} - 10\mathbf{k}$ and contains the point $(3, -6, 8)$. Find the vector equation for the plane Π in the form $\mathbf{r} \cdot \mathbf{n} = p$.

Question 7. Find the acute angle between the following pairs of planes:

a) $\Pi_1 : \mathbf{r} \cdot (-\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}) = 9, \quad \Pi_2 : \mathbf{r} \cdot (3\mathbf{i} + \mathbf{j} + 10\mathbf{k}) = 2$

b) $\Pi_1 : \mathbf{r} \cdot (-5\mathbf{i} - \mathbf{j} - 9\mathbf{k}) = 1, \quad \Pi_2 : \mathbf{r} \cdot (16\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}) = 13$

Question 8. It is given that the plane Π_1 has equation $\mathbf{r} \cdot (3\mathbf{i} - 2\mathbf{j} - 10\mathbf{k})$ and the plane Π_2 has equation $\mathbf{r} \cdot \mathbf{k} = 0$. Find the acute angle between Π_1 and Π_2.

Question 9. It is given that the plane Π_1 is perpendicular to the vector $8\mathbf{i} - \mathbf{j} + \mathbf{k}$ and contains the point $(2, 3, 1)$. It is also given that the plane Π_2 is perpendicular to the vector $\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$ and contains the point $(3, 9, 2)$.

a) Find the vector equation for the plane Π_1 in the form $\mathbf{r} \cdot \mathbf{n} = p$.
b) Find the vector equation for the plane Π_2 in the form $\mathbf{r} \cdot \mathbf{n} = p$.
c) Find the exact value of the cosine of the acute angle between Π_1 and Π_2.

Question 10. A particle moves along a path relative to a fixed origin O with equation:

$$\mathbf{r} = (800t^2 + 540t - 99)\mathbf{i} + (300 - 20t)\mathbf{j} + (800t^2 + 620t - 133)\mathbf{k},$$

where t is the time in seconds and distances are measured in metres. A plane has equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = 0$.

a) Find the exact time when the particle hits the plane.

The particle has velocity (in ms^{-1}) given by the following equation:

$$\mathbf{v} = (1600t + 540)\mathbf{i} - 20\mathbf{j} + (1600t + 620)\mathbf{k}.$$

a) Find the direction of the particle when the particle hits the plane.
b) Find the acute angle at which the particle hits the plane, giving your answer in degrees to 1 decimal place.

Question 11. The points $A(150, 0, -200)$, $B(300, 50, -150)$ and $C(400, 150, -300)$ all lie in the plane Π.

a) Find the equation of the plane Π in the form $\mathbf{r} \cdot \mathbf{n} = p$.

b) Find the acute angle that the plane Π makes with the plane $z = 0$.

Equations of Planes: Vector Equations

Questions 1–12

Question 1. Find a vector equation of the planes that pass through these points:

a) $(1, 0, 1)$, $(2, 1, 0)$ and $(1, 2, 2)$

b) $(-1, 2, -3)$, $(3, 2, 3)$ and $(-1, -1, 2)$

c) $(0, 0, 0)$, $(3, 4, 1)$ and $(-9, -3, 7)$

Question 2. Determine whether each of these points lies on the plane with equation

$$\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} :$$

a) $(9, 18, -7)$

b) $(24, 23, -17)$

c) $(19, 5, -22)$

Question 3. It is given that $A(2, -3, 5)$, $B(-6, 2, 3)$ and $C(34, -23, 13)$ relative to a fixed origin O.

a) Find λ and μ such that $\overrightarrow{OC} = \lambda\overrightarrow{OA} + \mu\overrightarrow{OB}$.

b) Explain what your answer to part (a) tells you about the points O, A, B and C.

Question 4. The plane Π has equation

$$\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -9 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 7 \\ -14 \end{pmatrix}.$$

a) Show that the vector $35\mathbf{i} + 33\mathbf{j} + 24\mathbf{k}$ is a normal vector to the plane.

b) Hence find the equation of the plane Π in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers.

Question 5. Determine whether the line with equation $\mathbf{r} = 2\mathbf{i} + \lambda(-\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$ lies in the plane with equation $\mathbf{r} = \mathbf{j} + \alpha(\mathbf{i} - \mathbf{k}) + \beta(\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})$.

Question 6. Find the acute angle between the following pairs of planes:

a) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$, $\Pi_2 : \mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \delta \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$

b) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 6 \\ 2 \\ -5 \end{pmatrix} + \alpha \begin{pmatrix} 4 \\ 0 \\ 5 \end{pmatrix} + \beta \begin{pmatrix} 3 \\ -2 \\ 0 \end{pmatrix}$, $\Pi_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} + \delta \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix}$

Question 7. Write the equation of each of these planes in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers:

a) $\mathbf{r} = \begin{pmatrix}0\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\0\end{pmatrix} + \mu\begin{pmatrix}0\\1\\1\end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix}1\\2\\5\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\1\end{pmatrix} + \mu\begin{pmatrix}4\\3\\1\end{pmatrix}$

c) $\mathbf{r} = \begin{pmatrix}4\\2\\-5\end{pmatrix} + \lambda\begin{pmatrix}3\\2\\0\end{pmatrix} + \mu\begin{pmatrix}0\\-5\\6\end{pmatrix}$

d) $\mathbf{r} = \begin{pmatrix}-2\\-8\\10\end{pmatrix} + \lambda\begin{pmatrix}-2\\4\\-7\end{pmatrix} + \mu\begin{pmatrix}5\\0\\-4\end{pmatrix}$

Question 8. Write the equation of each of these planes in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers:

a) $\mathbf{r} = \mathbf{i} - 10\mathbf{j} - 2\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} + 5\mathbf{k}) + \mu(-3\mathbf{i} - 2\mathbf{j} + 9\mathbf{k})$

b) $\mathbf{r} = \lambda(2\mathbf{i} + 3\mathbf{j} - 15\mathbf{k}) + \mu(8\mathbf{i} + 7\mathbf{k})$

Question 9. Write the equation of each of these planes in the form $\mathbf{r} \cdot \mathbf{n} = p$:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$

b) $\mathbf{r} = -8\mathbf{j} + 16\mathbf{k} + \lambda(20\mathbf{i} - 8\mathbf{j} + 5\mathbf{k}) + \mu(15\mathbf{i} + 8\mathbf{j})$

Question 10. Three planes, Π_1, Π_2 and Π_3, have equations:

$$\Pi_1 : 2x - 3y + 8z = -2, \qquad \Pi_2 : \mathbf{r} \cdot \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} = 3, \qquad \Pi_3 : \mathbf{r} = \begin{pmatrix} 3 \\ -5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix}.$$

a) Find the acute angle between Π_1 and Π_2.

b) Find the acute angle between Π_1 and Π_3.

Question 11. A plane contains the points $A(0, 2, 6)$, $B(-3, 1, 0)$ and $C(3, 0, -4)$. Find the equation of the plane in the form $ax + by + cz + d = 0$, where a, b, c and d are integers.

Question 12. A plane contains the points $A(4, -5, 1)$, $B(2, 3, 12)$ and $C(6, -2, 10)$. Find the equation of the plane in the form $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$.

Geometrical Configuration of Planes

Questions 1–28

Question 1. Describe all three possible geometrical arrangements of two planes.

Question 2. Describe all eight possible geometrical arrangements of three planes. Ignore replica scenarios in your answer, e.g. $\{1\&2, 3\}$ is the same as $\{1\&3, 2\}$.

Question 3. It is given that Π_1, Π_2 and Π_3 are planes. It is also given that the planes are neither parallel to one another nor coincident. Describe all three possible geometrical arrangements of the three planes.

Question 4. It is given that Π_1, Π_2 and Π_3 are distinct planes. The system of equations representing the planes does not have a unique solution, but the equations are consistent. Describe the geometrical arrangement of the three planes.

Question 5. It is given that $A = \begin{pmatrix} 1 & 4 & -2 \\ 3 & 2 & 2 \\ 5 & -1 & 7 \end{pmatrix}$.

a) Find A^{-1}.

b) Use your answer to (a) to solve the simultaneous equations $x + 4y - 2z = 10$, $3x + 2y + 2z = 12$, $5x - y + 7z = 8$.

c) State the geometrical configuration of the three planes.

Question 6. Three planes have equations $\Pi_1 : 2x + 3y - z = 1$, $\Pi_2 : x + 2y + z = 2$ and $\Pi_3 : 4x - y - z = 3$. Find the unique point of intersection of the three planes.

Question 7. Three planes have equations $\Pi_1 : x+4y-2z = 15$, $\Pi_2 : -6x-7y+z = -40$ and $\Pi_3 : 2x+5y-2z = 22$. Find the unique point of intersection of the three planes.

Question 8. Three planes have equations $\Pi_1 : 4x + y + 9z = k$, $\Pi_2 : 3x + 2y - 5z = 8$ and $\Pi_3 : 4x + 3y - 10z = 3$, where k is a constant. Show that the three planes always meet at a unique point.

Question 9. Determine the geometrical configuration of the following trios of planes:

a) $\Pi_1 : 6x + 2y - 4z = 1$, $\Pi_2 : 5x + y + z = 3$, $\Pi_3 : 3x + y - 2z = 8$

b) $\Pi_1 : 7x - 2y - z = 2$, $\Pi_2 : 9x - 3y + 15z = 12$, $\Pi_3 : 3x - y + 5z = 4$

c) $\Pi_1 : 20x + 30y - 15z = -6$, $\Pi_2 : \frac{4}{3}x + 2y - z = -\frac{2}{5}$, $\Pi_3 : 14x + 21y - \frac{21}{2}z = -\frac{21}{5}$

Question 10. Three planes have equations $\Pi_1 : 11x - 3y + z = 2$, $\Pi_2 : x + 2z = -5$ and $\Pi_3 : 3x - y - z = 4$. Determine their geometrical configuration.

Question 11. Three planes have equations $\Pi_1 : 4x-21y+6z = 10$, $\Pi_2 : 8x+2y+4z = 3$ and $\Pi_3 : 10x+8y+4z = 7$. Determine their geometrical configuration.

Question 12. Three planes have equations $\Pi_1 : 5x+6y+26z = 6$, $\Pi_2 : 2x-3y-4z = 6$ and $\Pi_3 : x-3y-6z = 12$. Determine their geometrical configuration.

Question 13. Three planes have equations $\Pi_1 : 2x + 4y + z = k$, $\Pi_2 : x + 19y - 8z = 0$ and $\Pi_3 : -2x + 4y - 5z = 0$, where k is a constant. Describe their geometrical configuration when (a) $k = 0$, (b) $k \neq 0$.

Question 14. Three planes have equations $\Pi_1 : 2x + ky + kz = k$, $\Pi_2 : kx + 3y + kz = k$ and $\Pi_3 : 4x + 2y + 2z = k$, where k is a constant. It is given that the three planes do not meet at a unique point when $k = 1$ or $k = 3$. Describe their geometrical configuration when (a) $k = 1$, (b) $k = 3$.

Question 15. Three planes have equations $\Pi_1 : 2x+4y+5z = 23$, $\Pi_2 : 3x+ky+4z = 20$ and $\Pi_3 : -2x+4y+kz = 6$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration of the three planes it represents, fully justifying your answers.

Question 16. Three planes have equations $\Pi_1 : (k+1)x + 3y - 2z = 1$, $\Pi_2 : kx + 4y - \frac{1}{4}(k+8)z = 2$ and $\Pi_3 : 4y - z = 3$, where k is a real constant. Show that the planes always meet at a unique point.

Question 17. Three planes have equations $\Pi_1 : kx - z = 0$, $\Pi_2 : x + 3y + kz = 1$ and $\Pi_3 : x + 2y + kz = 0$, where k is a constant.

a) Show that the planes always meet at a unique point.

b) Find the coordinates of the point of intersection in terms of k.

Question 18. Three planes have equations $\Pi_1 : x+ky-4z = -19$, $\Pi_2 : -6x+4y-3z = -42$ and $\Pi_3 : kx+y+4z = 19$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration of the three planes it represents, fully justifying your answers.

Question 19. Three planes have equations $\Pi_1 : kx + 2y + 3kz = 2$, $\Pi_2 : 4x + ky - 6z = -2$ and $\Pi_3 : x - 6z = 0$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration the planes represent, fully justifying your answers.

Question 20. Three planes have equations $\Pi_1 : -5x + 3y + (k+3)z = 2$, $\Pi_2 : kx + 2ky - 4z = -5$ and $\Pi_3 : 2x - 9y + z = 4$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration the planes represent, fully justifying your answers.

Question 21. Three planes have equations $\Pi_1 : 4x - 2y + 3z = 17$, $\Pi_2 : (k+1)x + 3y + (k-2)z = 15$ and $\Pi_3 : 4x + (k-3)y + 5z = -65$, where k is a constant. It is given that the three planes do not meet at a unique point.

a) Find the possible values of k.

b) For each value of k found in (a), identify which configuration the planes represent, fully justifying your answers.

Question 22. Three planes have equations $\Pi_1 : 2kx + (k+3)y + kz = 3$, $\Pi_2 : x - y + 2z = p$ and $\Pi_3 : 3x + y - 2z = 1$, where k and p are constants. Find the values of k and p such that the planes form a sheaf.

Question 23. The three planes $\Pi_1 : 6x - y + kz = 33$, $\Pi_2 : 2x - 3y + 5z = 25$ and $\Pi_3 : 4x + 2y - 3z = p$, where k and p are constants, form a sheaf.

a) Find the value of k.

b) Find the values of m and n such that $m(6x - y) + n(2x - 3y) \equiv 4x + 2y$ for any values of x and y.

c) By setting $z = 0$ and using your result to (b), find the value of p.

Question 24. It is given that $A = \begin{pmatrix} 3 & -4 & 17 \\ 1 & 10 & 0 \\ 2 & k-1 & 3 \end{pmatrix}$, where k is a constant.

a) Find the values of k for which A^{-1} exists.

For the following plane questions use $\Pi_1 : 3x - 4y + 17z = 17$, $\Pi_2 : x + 10y = 0$ and $\Pi_3 : 2x + 13y + 3z = m$, where m is a constant.

a) Find the unique point of intersection of the three planes in terms of m.

b) For $\Pi_4 : 3x - 4y + 17z = n$, $\Pi_5 : x + 10y = 1$ and $\Pi_6 : 2x + 14y + 3z = 2$, where n is a constant, find n so the planes are consistent. Hence describe their geometrical configuration.

Question 25. Three planes have equations $\Pi_1 : 2x-3y+kz = p$, $\Pi_2 : 5x-3y+k^2z = k-\frac{\sqrt{3}}{2}$ and $\Pi_3 : 4x+12y-z = 8$, where k and p are constants. It is given that the planes do not meet at a unique point.

a) Find the possible exact values of k.

b) Given that the equations are inconsistent, find the possible exact values of p.

c) Hence describe the geometrical configuration of the three planes.

Question 26. It is given that $A = \begin{pmatrix} k-1 & 2-k & 1 \\ k+2 & 1 & -3 \\ k+1 & 2 & -5 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -8 & k+1 \\ 15 & -20 & 15 \\ 7 & -16 & 15 \end{pmatrix}$, where k is a constant.

a) Find AB in terms of k.

b) It is given that $k = 4$. Find AB and hence find A^{-1}.

c) Three planes have equations $\Pi_1 : 3x - 2y + z = 40$, $\Pi_2 : 6x + y - 3z = 100$ and $\Pi_3 : 5x + 2y - 5z = -1000$. Use a matrix method to solve the three equations, and hence describe the geometrical configuration of the three planes.

Question 27. It is given that $A = \begin{pmatrix} 7 & -1 & 2 \\ k & 4 & -2 \\ 1 & -6 & 3 \end{pmatrix}$.

a) Find the value of k such that A is singular.

b) Use a matrix method to solve $7\sqrt{x} - \sqrt{y} + 2\sqrt{z} = 49$, $-\frac{7}{9}\sqrt{x} + 4\sqrt{y} - 2\sqrt{z} = \frac{10}{3}$ and $\sqrt{x} - 6\sqrt{y} + 3\sqrt{z} = -6$.

Question 28. Determine the geometrical configuration of the following trios of planes:

a) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + a\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + b\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$; $\Pi_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + m\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + n\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$; $\Pi_3 : \mathbf{r} = \begin{pmatrix} 8 \\ -3 \\ 2 \end{pmatrix} + p\begin{pmatrix} -2 \\ 4 \\ -3 \end{pmatrix} + q\begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}$.

b) $\Pi_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} + a\begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} + b\begin{pmatrix} 3 \\ 9 \\ 5 \end{pmatrix}$; $\Pi_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 10 \\ -10 \end{pmatrix} + m\begin{pmatrix} 6 \\ 18 \\ 10 \end{pmatrix} + n\begin{pmatrix} 3 \\ 15 \\ -6 \end{pmatrix}$; $\Pi_3 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + p\begin{pmatrix} 3 \\ 9 \\ 4 \end{pmatrix} + q\begin{pmatrix} 2 \\ 10 \\ -4 \end{pmatrix}$.

Lines and Planes

Questions 1–25

Question 1. Find the acute angle between the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -9 \end{pmatrix}$ and each of the following planes:

a) $x = 0$
b) $y = 0$
c) $z = 0$

Question 2. Find the acute angle between the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix}$ and the plane with equation $4x + y - 2z = 3$.

Question 3. Determine the geometrical configuration of the line with equation $\mathbf{r} = \begin{pmatrix} 3 \\ 7 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -8 \\ 9 \end{pmatrix}$ and the plane $6x + 15y + 10z - 90 = 0$.

Question 4. Determine the geometrical configuration of the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ \frac{11}{3} \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -4 \\ -5 \end{pmatrix}$ and the plane $3x + 12y - 6z + 13 = 0$.

Question 5. Determine the geometrical configuration of the line with equation $\mathbf{r} = \begin{pmatrix} -1 \\ 5 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}$ and the plane $5x + 3y + 2z - 4 = 0$.

Question 6. A road can be modelled as a plane using the equation $x + y + 10z = 1$, where x, y and z are measured in metres. A gas pipe can be modelled as a straight line passing through the points $(1, 1, -0.5)$ and $(0.2, 0.4, -0.6)$.

a) Find the acute angle that the line makes with the plane.

An engineer must connect a monitoring device to the pipe from a point on the road, P. It is given that P has coordinates $(0.8, 92.1, -9.19)$.

a) Find the shortest distance from the point P to the pipe.

Question 7. Find the coordinates of where the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \\ -2 \end{pmatrix}$ intersects the following planes:

a) $x = 0$
b) $y = 0$
c) $z = 0$

Question 8. Find the coordinates of where the line with equation $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$ intersects the plane with equation $4x + y - z = 9$.

Question 9. Point A has coordinates $(3, 2, 4)$. Find the exact coordinates of the image of point A when it is reflected in the plane with equation $x + y - z = 0$.

Question 10. Point P has coordinates $(5, -2, \frac{1}{2})$. Find the exact coordinates of the image of point P when it is reflected in the plane with equation $10x - y - 2z = -54$.

Question 11. It is given that l_1 intersects the plane Π at the point P, and l_2 intersects the plane Π at the point Q, where

$$l_1 : \ \mathbf{r} = \begin{pmatrix} 2 \\ 8 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad \Pi : 9x - 20y + 2z = 6.$$

Find the exact length of PQ.

Question 12. The line l has the equation $\mathbf{r} = \begin{pmatrix} 2 \\ a \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} b \\ 4 \\ 1 \end{pmatrix}$ and the plane Π has the equation $2x + y + z = 5$.

a) Given that the line lies on the plane Π, find the values of a and b.

b) Given instead that the line l intersects the plane Π at precisely one point, state the condition on a and b for this to be the case.

Question 13. The line l has equation $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ and the point P has coordinates $(-3, 7, 5)$.

a) Verify that P lies on the line l.

It is given that the plane Π is perpendicular to the line l and contains the point P.

a) Find the cartesian equation of the plane Π.

The plane Π contains the point $Q\left(-3, 7 - \frac{1}{\sqrt{5}}, 5 + \frac{2}{\sqrt{5}}\right)$. The point $R(a, b, c)$ lies on the line l such that the distance QR is 5 units.

a) Find the possible coordinates of R.

Question 14. Points A and B have coordinates $(3, -2, 3)$ and $(4, -1, -2)$ respectively and lie on the line l_1.

a) Find the equation of l_1 in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$.

The points A and B lie on the plane Π with equation $px + qy + rz + d = 0$, where p, q, r and d are constants. The point C with coordinates $(1, 2, 1)$ also lies on Π.

a) Find the equation of l_2, the line that passes through A and C, in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$.

b) Show that $\mathbf{n} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ is perpendicular to both lines l_1 and l_2.

c) Hence find the equation of the plane Π.

Points D and E have coordinates $(8, 5, 9)$ and $(6, 4, 10)$ respectively and lie on the line l_3. The line l_3 intersects the plane Π at the point P.

a) Find the exact coordinates of the point P.

b) Find the acute angle that is made when l_3 intersects the plane Π.

It is given that the point F lies on l_3 such that $PF = 1$ unit.

a) Find the possible coordinates of F.

Question 15. Plane Π has equation $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -9 \\ 2 \end{pmatrix} = 18$. Determine whether the following lines intersect at a single point, or whether they lie within Π, or whether they are parallel to Π. If they intersect the plane, determine the coordinates of this point of intersection:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} 6 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 13 \\ 48 \end{pmatrix}$

c) $\frac{x-6}{2} = \frac{y-4}{4} = \frac{z-18}{15}$

d) $x + 8 = 3 - y = \frac{z+1}{6}$

Question 16. Two lines l_1 and l_2 are described as follows:

$$l_1 : \ \frac{x+5}{3} = y - 1 = 2 - z, \qquad l_2 : \ 5 - x = y + 3 = \frac{z-5}{2}.$$

a) Show that l_1 and l_2 are skew lines.

b) Find a vector that is perpendicular to both l_1 and l_2.

c) Find a cartesian equation of the plane which is parallel to l_1 and contains l_2.

d) Find a cartesian equation of the plane which is parallel to l_2 and contains l_1.

The line l_3 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ 8 \end{pmatrix}$. The line l_3 intersects the plane found in (c) at the point P and the plane found in (d) at the point Q.

a) Find the exact distance PQ.

Question 17. Find the coordinates of where the line with equation $2x + 8 = \frac{4-y}{3}$, $z = 2$ intersects the plane with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ 3 \\ 5 \end{pmatrix}$.

Question 18. One face of a mountainside can be modelled by a plane that passes through the points $(60, 70, 320)$, $(50, 120, 235)$ and $(130, 100, 155)$, where measurements are in metres.

a) Find a vector equation of the plane in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$.
b) Show that the vector $3\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ is perpendicular to the plane.
c) Hence write down the cartesian equation of the plane.
d) A tunnel is to be dug into the mountainside that is perpendicular to the rock face, so that it passes through the point $(20, 65, 100)$. Find the coordinates of the point where the tunnel should begin and find the exact length of the tunnel so that the length of the tunnel is minimised.

Question 19. It is given that

$$\Pi_1 : 4x - y - z = 3, \qquad \Pi_2 : 3x + 2y + 2z = 9.$$

a) Show that Π_1 and Π_2 are not perpendicular.
b) Find a point that lies on both Π_1 and Π_2.
c) Find a vector that is parallel to both Π_1 and Π_2.

It is given that the plane Π_3 is perpendicular to both Π_1 and Π_2.

a) Find the equation of the plane Π_3 that passes through the point $(5, 3, 1)$.

Question 20. It is given that

$$\Pi : 5x + 8y + 9z = 16, \qquad l : \ \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}.$$

The points P and Q lie on l so that they are both 14 units from Π.

a) Find the coordinates of P and Q.

It is given that the point A has coordinates $(11, -6, 1)$ and that A lies on Π.

a) Find the exact area of the triangle APQ.

Question 21. It is given that

$$\Pi : x - y + z = 6, \qquad l : \ \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -7 \\ 6 \end{pmatrix}.$$

Find an equation of the line that is the reflection of the line l in the plane Π.

Question 22. It is given that

$$\Pi_1 : 6x + y - 2z = 8, \qquad \Pi_2 : 5x - 3y - 10z = 2.$$

Show that the line l with equation

$$\mathbf{r} = \begin{pmatrix} 22 \\ -64 \\ 30 \end{pmatrix} + \lambda \begin{pmatrix} 16 \\ -50 \\ 23 \end{pmatrix}$$

lies on both planes.

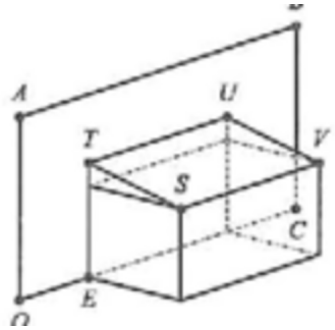
Question 23. Find the equation of the line of intersection between the planes:

$$\Pi_1 : 2x - y + z = 5, \qquad \Pi_2 : 3x + y - z = 2.$$

Question 24. Find the equation of the line of intersection between the planes:

$$\Pi_1 : z = 0, \qquad \Pi_2 : 5x + 2y + 3z = 8.$$

Question 25. The diagram shows a rabbit hutch built against a wall.



The wall $OABC$ can be modelled by the plane with equation $x = 0$. The rabbit hutch $STUV$ can be modelled by the plane with equation $x - 4z + 20 = 0$.

a) The lid of the rabbit hutch is in contact with the wall along the line TU. Find the equation of the line passing through T and U in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.

b) Find the acute angle ETS.

Shortest Distances in 3D: Points and Lines

Questions 1–33

Question 1. Find the shortest distance from the point $(2, 5, -3)$ to the x-axis.

Question 2. Find the shortest distance from the point $(1, 8, -9)$ to the z-axis.

Question 3. The point P has position vector $\overrightarrow{OP} = 6\mathbf{i} + 12\mathbf{j} - 5\mathbf{k}$. The line l passes through the point P and is perpendicular to the vector $\overrightarrow{OP}$. Find the shortest distance between l and the origin O.

Question 4. Find the shortest distance from the origin to the line with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$.

Question 5. Find the shortest distance from the point $(2, 1, 1)$ to the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$.

Question 6. A sphere has centre $(-4, 3, 8)$ and radius r. Its surface touches the line that passes through $(1, 0, 1)$, parallel to the vector $\mathbf{i} + \mathbf{j}$. Find r.

Question 7. Find the shortest distance from $(1, 0, 1)$ to the line with equation $\frac{x-1}{2} = y = z + 2$.

Question 8. Find the shortest distance from $(6, 0, 4)$ to the line with equation $x = 2y = 3 - z$.

Question 9. It is given that the points A and B have coordinates $(2, -1, 1)$ and $(9, 12, 4)$ respectively. The line l passes through the points A and B. The point P has coordinates $(9, -22, 0)$.

a) Show that P does not lie on the line l.

b) It is given that Q is the point on l that is closest to point P. Find the coordinates of the point Q.

c) Hence find the exact shortest distance between the point P and the line l.

Question 10. Given that the points $A(3, -1, 0)$ and $B(5, 1, 2)$ lie on the plane $\Pi : 2x + 7y - 9z = -1$, find the equation of the line that lies on the plane Π such that every point on the line is equidistant from A and B.

Question 11. The points P and Q have coordinates $(2, 2, 8)$ and $(10, 3, 0)$ respectively.

a) Show that P and Q lie on the plane Π with equation $3x - 8y + 2z - 6 = 0$.

b) Find the equation of the line l that lies on the plane Π such that every point on the line is equidistant from P and Q.

c) The point R lies on the line l such that the chord $PR = \frac{43}{\sqrt{6}}$ units. Find the possible coordinates of R.

Question 12. The points A, B and C have coordinates $(2, 1, 2)$, $(3, 4, 1)$ and $(0, 1, 1)$ respectively.

a) Find the cartesian equation of the plane Π that contains A, B and C.

b) Find the equation of the line l_1 contained by Π such that every point on l_1 is equidistant from A and B.

c) Find the equation of the line l_2 contained by Π such that every point on l_2 is equidistant from B and C.

d) It is given that P is the point of intersection of l_1 and l_2. Find the coordinates of P.

Question 13. An underground railway tunnel can be modelled by a straight line given by the equation

$$\mathbf{r} = \begin{pmatrix} 50 \\ -765 \\ -116 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}.$$

The ground above can be modelled by a plane given by the equation $x - y + 6z = 300$. An emergency exit from the tunnel, P, is to be built to connect to a point $Q(10, k, 10)$ on the ground.

a) Find the value of k.
b) Find the coordinates of P such that the distance from Q to P is minimised.
c) Hence find the exact distance from Q to P.

Question 14. It is given that the line l has equation $\mathbf{r} = \mathbf{i} + \mathbf{j} + 2\mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + \mathbf{k})$ and the point P has coordinates $(k, 2, -1)$.

a) Show that the shortest distance between P and the line l, in terms of k, is $\frac{1}{3}\sqrt{6k^2 + 12k + 24}$.
b) Hence find the value of k that gives the minimum value of $\frac{1}{3}\sqrt{6k^2 + 12k + 24}$. State this minimum value.

Question 15. The point A has coordinates $(1, 2, 1)$. Find the coordinates of the image of the point A after it is reflected in the line with equation $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$.

Question 16. It is given that the point A has coordinates $\left(29, 29, -\frac{109}{3}\right)$. Find the coordinates of the image of the point A after it is reflected in the line with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix}$.

Question 17. Find the shortest distance between the following pair of parallel lines:

$$l_1 : \ \mathbf{r} = \mathbf{i} + \lambda(\mathbf{j} + \mathbf{k}), \qquad l_2 : \ \mathbf{r} = \mathbf{k} + \mu(\mathbf{j} + \mathbf{k}).$$

Question 18. Find the shortest distance between the following pair of parallel lines:

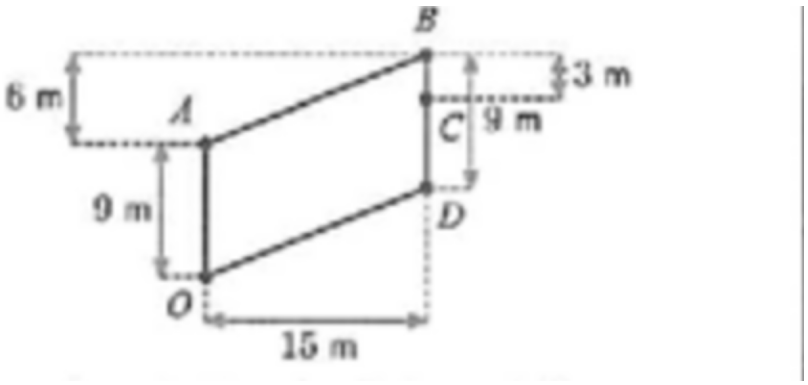
$$l_1 : \ \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Question 19. Find the shortest distance between the lines with equations:

$$l_1 : \ \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}.$$

Question 20. Two crane arms are to be used in a movie stunt. The crane arms are modelled as straight lines using coordinates in metres. One crane arm has its base at $(20, 20, 60)$ and its end at $(80, 80, 90)$. The other crane arm has its base at $(60, 20, 70)$ and its end at $(30, 70, 100)$. Find the shortest distance between the two crane arms.

Question 21. A drone light show involves two drones that are programmed to follow straight paths at different angles. The drones are programmed to remain within the parallelogram shown in the diagram below, as seen from above. Take the point O to have coordinates $(0, 0, 0)$. Drone 1 starts from 18 metres above point A and follows a straight path to reach a height of 4 metres above point D. Meanwhile, Drone 2 starts from 3 metres above point B and follows a straight path to reach a height of 4 metres above point O, before then immediately changing direction and following a straight path to reach a height of 20 metres above point C.



Determine whether the flight path for Drone 2 is closer to the flight path of Drone 1 when it is travelling from above B to above D, or when it is travelling from above O to above C.

Question 22. Two ski lifts at a resort are to be built so that they cross over one another. The first is to pass between the two points $(-180, -20, 25)$ and $(260, 2, 36)$. The second is to pass between the two points $(-220, 44, -1)$ and $(290, 10, 50)$. All measurements are given in metres, where the origin is the reception chalet. For safety, the two ski lifts cannot pass within 10 metres of one another.

a) Determine whether the design of the two ski lifts passes the safety regulation.

b) State any assumptions that have been made that may affect the design passing the safety regulations.

Question 23. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 2 \\ 1 \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix},$$

where k is a constant. It is also given that the shortest distance between the lines l_1 and l_2 is 2 units. Find the value of k.

Question 24. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ k \\ k \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix},$$

where k is a constant. It is also given that the shortest distance between the lines l_1 and l_2 is 1 unit. Find the possible values of k.

Question 25. It is given that

$$l_1 : \ \mathbf{r} = \begin{pmatrix} -1 \\ k \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 2 \\ -1 \end{pmatrix}, \qquad l_2 : \ \mathbf{r} = \begin{pmatrix} k \\ k \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 3k \\ -2 \\ 2 \end{pmatrix},$$

where k is a positive constant. It is also given that the shortest distance between the lines l_1 and l_2 is 1 unit. Find the value of k.

Question 26. Two cables are hung across a room near the ceiling. The cables are modelled as straight lines using coordinates in metres, relative to a fixed origin O. One cable runs from the point $(3.5, k, 2.5)$ to the point $(1.5, 0.5, 3)$. The other cable runs from the point $(1.5, 1, 3)$ to the point $(4, 1.5, 2.5)$. The two cables are to be attached at their closest point by a wire that is 40 cm in length. Find the possible values of k.

Question 27. Find the shortest distance between the lines with equations, giving your answer to 3 significant figures:

$$l_1: \ \frac{x+2}{4} = \frac{y-5}{2} = \frac{z+7}{3}, \qquad l_2: \ \frac{x-1}{8} = \frac{y+2}{4} = \frac{z-4}{5}.$$

Question 28. Find the exact shortest distance between the lines with equations:

$$l_1: \ x-1 = \frac{y+3}{5} = \frac{z}{2}, \qquad l_2: \ \frac{x+5}{2} = y-3 = 1-z.$$

Question 29. Find the exact shortest distance between the lines with equations:

$$l_1 : \ 2x + 5 = 4 - y = 5 - 2z, \qquad l_2 : \ \frac{x}{2} = 4y = z - 3.$$

Question 30. It is given that the shortest distance between the following lines is $\sqrt{2}$ units:

$$l_1 : \ x - 1 = 2 - y = 4 - z, \qquad l_2 : \ x = 1, \ ky = z,$$

where k is a constant. Find the value of k.

Question 31. It is given that two lines have equations:

$$l_1 : \left(\mathbf{r} - \begin{pmatrix} 7 \\ 6 \\ 10 \end{pmatrix}\right) \times \begin{pmatrix} -6 \\ 5 \\ 3 \end{pmatrix} = \mathbf{0}, \qquad l_2 : \left(\mathbf{r} - \begin{pmatrix} 0 \\ 2 \\ -9 \end{pmatrix}\right) \times \begin{pmatrix} 6 \\ -5 \\ -3 \end{pmatrix} = \mathbf{0}.$$

a) Explain how you know that the two lines are parallel.
b) Find the exact distance between the two lines.

Question 32. Find the exact shortest distance between the lines with equations:

$$l_1 : \left(\mathbf{r} - \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} = \mathbf{0}, \qquad l_2 : \left(\mathbf{r} - \begin{pmatrix} -1 \\ 5 \\ -3 \end{pmatrix}\right) \times \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} = \mathbf{0}.$$

Question 33. Find the exact shortest distance between the lines with equations:

$$l_1 : \ 2x + 3 = 3 - z, \ y = 2, \qquad l_2 : \ \left(\mathbf{r} - \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right) \times \begin{pmatrix} -7 \\ 2 \\ 3 \end{pmatrix} = \mathbf{0}.$$

Shortest Distances in 3D: Planes

Questions 1–21

Question 1. Find the shortest distance from the origin to the plane with equation $x + 2y + 9z = 344$.

Question 2. Find the shortest distance between the point $A(2, 1, 2)$ and the plane with equation $x + y + z = 3$.

Question 3. Find the shortest distance from the given point P and the given plane Π:

a) $P(10, 3, 1)$, $\Pi : x - y - z = 3$

b) $P(-3, 2, 8)$, $\Pi : 2x + 5y - 8z = 6$

c) $P(-9, 0, 60)$, $\Pi : -5x - 3y + 9z = 10$

d) $P(6, 6, -1)$, $\Pi : -y - 2z = 1$

Question 4. It is given that the point A has position vector $-10\mathbf{i} + 20\mathbf{j} - 50\mathbf{k}$. Find the exact distance from the point A to the plane with equation $4x + y - 15z = 1698$.

Question 5. It is given that the lines l_1 and l_2 and the plane Π have equations:

$$l_1 : \mathbf{r} = \begin{pmatrix} 10 \\ 5 \\ -10 \end{pmatrix} + \lambda \begin{pmatrix} 30 \\ 1 \\ 5 \end{pmatrix}, \qquad l_2 : \mathbf{r} = \begin{pmatrix} 20 \\ 5 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -80 \\ -3 \\ 5 \end{pmatrix}, \qquad \Pi : x - 2y + 3z - 218 = 0.$$

The coordinates of where l_1 and l_2 intersect is point P.

a) Find the coordinates of point P.

b) Find the shortest distance of P from the plane Π.

Question 6. It is given that the plane Π has equation $3x - 2y - 14z + 39 = 0$ and that the line l has equation

$$\mathbf{r} = \begin{pmatrix} 5 \\ 28 \\ 18 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}.$$

The point Q lies on the line l such that the shortest distance between Q and the plane Π is $\sqrt{209}$ units. Find the possible coordinates of Q.

Question 7. A large flat area of open ground is modelled as a plane with equation $z = 0$. A slope mine is dug due to a coal bed, which can be modelled as a plane with equation $\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 50 \end{pmatrix} = -4935$, where measurements are in metres. The entrance to the slope mine has coordinates $(25, 25, 0)$, relative to a fixed origin.

a) Find the shortest distance between the entrance to the slope mine and the coal bed.

Due to the shortest distance being a route that was too steep, it is instead decided to reach the coal bed via a tunnel that can be modelled by the equation

$$\mathbf{r} = \begin{pmatrix} 25 \\ 25 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ 11/7 \\ 1 \end{pmatrix}.$$

b) Find the distance from the tunnel entrance to where the tunnel meets the coal bed.

Question 8. It is given that the points A, B and C have coordinates $(2, -1, 0)$, $(0, -2, -1)$ and $(2, 2, -2)$ respectively. It is also given that θ is the acute angle between the lines BA and BC. The triangle ABC lies in the plane Π.

a) Show that $\sin\theta = \frac{\sqrt{22}}{6}$.
b) Find the equation of the plane Π.
c) The point D has coordinates $(3, 1, 5)$. Show that D does not lie in the plane Π.
d) Find the shortest distance between D and the plane Π.
e) Hence find the exact volume of the tetrahedron $ABCD$, given that the volume is equal to $\frac{1}{3}\times$ area of base $\times$ perpendicular distance from base to apex.

Question 9.

a) Find the shortest distance from the origin to the plane Π with equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = 5$.

b) Show that the point A with coordinates $(-13, 4, 3)$ lies on the plane Π.

c) Write the equation of the plane Π in the form $\mathbf{r} \cdot \hat{\mathbf{n}} = d$, where $\hat{\mathbf{n}}$ is the unit vector in the direction of $\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$.

Question 10. Jake is trying to find the shortest distance from the point A with coordinates $(2, -3, 4)$ to the plane Π_1 with equation $\mathbf{r} \cdot \begin{pmatrix} -2 \\ 4 \\ 5 \end{pmatrix} = 8$. First, Jake finds the equation of Π_2, the plane that is parallel to Π_1 and contains A.

a) Find the equation of Π_2 in the form $\mathbf{r} \cdot \hat{\mathbf{n}} = d$.

b) Find Π_1 and Π_2 in the form $\mathbf{r} \cdot \hat{\mathbf{n}} = d$.

c) Show how Jake can use his two answers from part (b) to find the shortest distance from A to the plane Π_1.

Question 11. Find the shortest distance between the point $A(2, -10, -642)$ and the plane with equation $\mathbf{r} \cdot \begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} = 2$.

Question 12. Find the shortest distance from the origin to the plane with equation

$$\mathbf{r} = \begin{pmatrix} 10 \\ 66 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 3 \\ 9 \end{pmatrix}.$$

Question 13. Find the shortest distance from the point with coordinates $(2, 1, 7)$ to the plane with equation

$$\mathbf{r} = \begin{pmatrix} 120 \\ -15 \\ 985 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}.$$

Question 14. The plane Π contains both the point with coordinates $(2, 3, -5)$ and the line with equation

$$\mathbf{r} = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}.$$

Find the shortest distance from the origin to the plane Π.

Question 15. Find the distance between the parallel planes $x + y + z = 40$ and $x + y + z = 60$.

Question 16. An artist is designing a sculpture made from several parallel pieces of acrylic. Two of the acrylic pieces can be modelled by the planes given by

$$\Pi_1 : px + 2y + qz = 300, \qquad \Pi_2 : px + 2y + qz = 627.$$

It is given that the acrylic piece modelled by Π_1 must pass through the points with coordinates $(100, -150, -4)$ and $(150, -100, 10)$.

a) Find the values of p and q.

b) Find the distance between the planes Π_1 and Π_2.

Question 17. The planes Π_1 and Π_2 are parallel. Given that Π_1 has equation $x + 2y + 2z = 10$ and that the distance between Π_1 and Π_2 is 5 units, find the two possible equations of plane Π_2.

Question 18. A laser is set up between two large parallel mirrors where distances are in metres. Given that $p > 0$, the mirrors can be modelled by the equations

$$\Pi_1 : px + 9y + 3z = 6, \qquad \Pi_2 : px + 9y + 3z = 26.$$

When fired, the laser passes through the points with coordinates $(4, 2, 0)$ and $(4, 10, 8)$. The laser first hits Π_1, making an acute angle θ with the normal to Π_1 such that $\sin\theta = \frac{\sqrt{6}}{3}$.

a) Find the value of p.
b) Find the coordinates of where the laser first hits Π_1.
c) Find the distance between the two mirrors.

Question 19. Find the distance between the line with equation

$$\mathbf{r} = \begin{pmatrix} 2 \\ 8 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}$$

and the plane with equation $-31x + 14y + 13z + 2615 = 0$.

Question 20. Find the distance between the line with equation $2x = \frac{y-3}{2} = \frac{60-6z}{13}$ and the plane with equation

$$\mathbf{r} \cdot \begin{pmatrix} 3 \\ -4 \\ -3 \end{pmatrix} = 60.$$

Question 21. It is given that the point $P(2, 1, 2)$ is the same distance from the plane $4x + y + 8z = -65$ as it is from the line

$$\mathbf{r} = \begin{pmatrix} 0 \\ -13 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 3 \\ k \end{pmatrix}.$$

Find the value of k.